

Fair Interpersonal Comparisons

Characterizing Contour-Set Volume Measures of Well-Being

Craig Pesme

Paris School of Economics

Abstract

How can welfare be compared across individuals with heterogeneous preferences without imposing arbitrary interpersonal utility normalizations? A situation is a pair (x, R) , consisting of a bundle and a preference relation. Building on the contour-based approach to fairness and well-being measurement, the paper first characterizes the relevant information in a situation by its *opportunity-equivalent profile*: the collection of indices of nested opportunity sets whose best attainable element is just indifferent to the individual's actual bundle. The main question of the paper is then whether such contour-based well-being measures can be neutral with respect to commodity labels and units of measurement. It is shown that these neutrality requirements are incompatible with Pareto on unrestricted domains of continuous, convex, and monotone preferences. Positive results therefore require common restrictions on the geometry of admissible preferences. Under such restrictions, and conditional on a separability axiom of contour information, neutrality characterizes a family of volume-type well-being measures. These measures evaluate a situation by the weighted volume of the relevant lower or upper contour set. They can therefore be interpreted as a continuous generalization of Borda-count ranking methods: they count the alternatives dominated by the individual's actual bundle. For CES preferences, the representation yields a geometrically normalized utility index.

Keywords: Interpersonal comparisons, Fairness, Opportunities.

JEL codes: D63, I31, D71, C43.

1 Introduction

Evaluating public policies requires comparing their effects on people’s quality of life. Measuring these effects is relatively straightforward when well-being can be represented along a single dimension, such as health or income. A similar reduction is possible in multidimensional environments if all individuals are assumed to share the same preferences over bundles. In that case, one can construct a common welfare index, aligned with the shared preference ordering, and use it to compare individuals’ situations. This is the logic behind the concept of real income in economies with multiple market commodities: nominal resources are converted into a common measure of purchasing power by correcting for prices and preferences.

Though extremely useful, the common-preference assumption is also highly restrictive. When individuals have heterogeneous tastes, their ordinal preferences rank their own bundles, but they do not by themselves say how one person’s attained preference level should be compared with another’s. A utility representation of each individual’s preferences would provide a numerical scale, but these scales are arbitrary up to increasing transformations and therefore cannot be compared without further normalization. The question addressed in this paper is how to construct interpersonal well-being comparisons that respect heterogeneous preferences while avoiding arbitrary interpersonal utility normalizations.

This question lies at the intersection of several literatures. The problem of interpersonal comparability has been central in social choice and welfare economics since the Bergson-Samuelson formulation of social welfare and Arrow’s critique of unrestricted ordinal aggregation (Bergson 1938; Samuelson 1947; Arrow 1963). A large axiomatic literature has studied which informational assumptions about utilities make social evaluation possible, and how different degrees of interpersonal comparability affect admissible social welfare functionals (d’Aspremont and Gevers 1977; d’Aspremont and Gevers 2002; Bossert and Weymark 2004; Fleurbaey and Tadenuma 2014; Piacquadio 2017). The present paper adopts the common resource-based approach, which considers that what matters for comparing individuals is their resources.

This resource-based perspective connects the paper to theories of equality of resources and fair allocation. The normative idea that distributive justice should be formulated in terms of fair claims on resources is central in both the philosophical and economic literatures on justice (Rawls 1971; Sen 1979; Dworkin 1981). In economics, this idea has been developed through a large axiomatic literature on fair division, public goods, and cost sharing. One branch of this literature studies fair allocation rules, asking which allocation should be selected for each economy (Moulin 1987; Moulin 1996; Moulin and Thomson 1997; Thomson 2011). Another branch studies fair social orderings, asking how all feasible allocations should be ranked (Fleurbaey and Maniquet 2008; Fleurbaey and Maniquet 2011). Both approaches typically address two questions simultaneously: how individual welfare should be measured, and how individual welfare levels should be aggregated into a social evaluation.

The present paper does not focus on choosing a social aggregator over individual situations. Instead, it focuses on the prior question of how to assign an individual well-being index to a pair consisting of a consumed bundle and a preference relation. The immediate starting point of this paper is Fleurbaey and Maniquet (2017), who ask the exact same question.

Their framework evaluates a situation (x, R) , where x is a bundle and R is the individual's preference relation. They impose Pareto, requiring the well-being measure to respect each individual's own preferences, and introduce Nested Contour, a fairness condition requiring that one situation be ranked above another when the upper contour set of the former is disjoint from the lower contour set of the latter. They show that, under continuity and Pareto, this is equivalent to an informational axiom: well-being at a situation depends only on the indifference contour attained in that situation. They then study two strengthenings of this contour requirement. One leads to well-being measures based on Samuelson's money-metric utility (Samuelson 1974; Samuelson and Swamy 1974), which asks what expenditure is required, at a reference price vector, to attain a given utility level. The other leads to measures based on the dual idea of ray (or distance) utility functions, which measure well-being by the proportion of a reference bundle that is indifferent to the actual bundle (Samuelson 1977; Deaton 1979; Pazner 1979). The lattice structure of indifference contours explains the duality between these two families. These constructions are appealing because they are ordinal in individual preferences: they do not require interpersonal comparisons of utility representations. Yet they replace the utility-normalization problem with a reference-object problem. Money metrics require a reference price vector; ray utilities require a reference ray. Different choices can generate different interpersonal rankings, some potentially leading to undesirable unfair social recommendations (Blackorby and Donaldson 1988).

The present paper takes the same resource-based perspective and starts from the same core axioms - in particular Nested Contour. The first contribution is a representation theorem for such contour-based well-being measurement. An opportunity mapping is an ordered family of compact, free-disposal opportunity sets indexed by a scalar (Thomson 1994). For each such mapping, the opportunity-equivalent value of a situation is the index at which the individual can reach exactly her actual indifference contour. Taking all opportunity mappings together gives an opportunity-equivalent profile. The theorem shows that every continuous, preference-respecting well-being function satisfying Nested Contour can, on the relevant contour domain, be written as a monotone aggregation of this profile. Moreover, ray profiles alone already contain the same information: knowing where every positive ray intersects the attained indifference contour is enough to reconstruct the contour whenever the contour is determined by its strictly positive part. This result clarifies the link between ray utilities and other measures such as money-metrics that are recurrent throughout welfare economics and fair allocation.

The paper then asks which aggregations of such opportunity-equivalent information can satisfy demanding neutrality requirements. This shift brings the analysis into contact with another literature: the axiomatic ranking of opportunity sets in terms of freedom of choice. Following Sen's emphasis on capabilities and the value of opportunities, many contributions study how opportunity sets should be compared independently of the particular option ultimately chosen (Sen 1991; Sen 1992). Early work often focuses on finite opportunity sets and asks whether freedom should be measured by cardinality, diversity, preference satisfaction, or other features of the available set (Jones and Sugden 1982; Pattanaik and Xu 1990; Bossert, Pattanaik, and Xu 1994; Arrow 1995; Puppe 1996; Gravel 1998; Pattanaik and Xu 1998; Pattanaik and Xu 2000a; Sugden 1998). Pattanaik and Xu extend this program to economic environments with compact opportunity sets in Euclidean commodity spaces (Pattanaik

and Xu 2000b). In particular, Xu (2004) focuses on linear budget sets in the non-negative orthant and characterizes the volume ranking by three axioms: Symmetry, Monotonicity, and Invariance of Scaling Effects. The present paper borrows directly from this opportunity-set literature, and especially from Xu (2004), but it changes the object and the interpretation. In Xu’s analysis, the budget set is the primitive object, preferences are deliberately absent, and the goal is to measure the intrinsic freedom offered by a linear opportunity set. Symmetry says that commodity labels should not matter; Monotonicity says that a strict set expansion increases freedom; Invariance of Scaling Effects says that the ranking should not change when the unit of measurement of a commodity is rescaled. Here, the primitive object is not an exogenous opportunity set but a preference-based situation. The relevant sets are the lower and upper contour sets generated endogenously by heterogeneous preferences at the consumed bundle. Symmetry and scaling invariance are adapted to our framework: a well-being function should be invariant to the labels assigned to goods and to their units of measurement. Pareto and Nested Contour play the role of preference-sensitive monotonicity.

The first result is negative. On the unrestricted domain of continuous, convex, and weakly monotone preferences, Pareto, Symmetry, and Unit Invariance are jointly inconsistent. This impossibility is not merely a boundary pathology generated by preferences that ignore some goods: it persists even on domains whose preferences are strongly monotone in the interior. Thus, unlike Xu’s exogenous-budget problem, where symmetry, monotonicity, and scale invariance select volume, the preference-sensitive problem first reveals the need for domain restrictions. Neutral interpersonal comparisons can exist only when admissible preferences share enough asymptotic structure near the boundary of the commodity space and at infinity. In geometric terms, their indifference sets must have a common tail behavior. Under an additional additive separability condition, this common geometry leads to volume-type indices: well-being is represented by the weighted volume of either the lower contour set or the upper contour set. These measures can be interpreted as counting, with a symmetric and unit-invariant density, the alternatives dominated by the individual’s actual bundle, or the alternatives that dominate it.

The main characterization therefore splits according to how indifference contours behave near the boundary of the commodity space. Some preferences have contours that approach the axes fast enough for the set of bundles worse than the actual bundle to have finite weighted volume. For these preferences, well-being is represented by the weighted volume of the lower contour set. Other preferences have contours that approach the axes slowly enough for the set of bundles better than the actual bundle to have finite weighted volume, when measured with a negative volume parameter. For these preferences, well-being is represented by the weighted volume of the upper contour set, with boundary bundles assigned the bottom value. Thus the relevant volume may be either the volume of dominated alternatives or the volume of dominating alternatives, depending on the common tail geometry of the admissible preference domain. This characterization extends the opportunity-set volume logic of Xu (2004). When preferences are perfect substitutes, lower contour sets are linear budget sets, and the ordinary-volume member of the family reproduces Xu’s volume ranking. In this sense, the paper provides a preference-sensitive analogue of Xu’s theorem: the opportunity set to be measured is not given exogenously, but is the contour set induced by the individual’s own preferences at her actual bundle.

Finally, the characterization also connects to recent work on welfare and price indices with heterogeneous preferences. Money-metric and cost-of-living approaches remain central in empirical welfare measurement, especially when households face different prices and have heterogeneous and non-homothetic preferences (Slesnick 1998; Fleurbaey 2009; Jaravel 2021; Baqaee and A. Burstein 2023; Jaravel and Lashkari 2024; Baqaee, A. T. Burstein, and Koike-Mori 2024). In the CES case with a common elasticity of substitution, the characterized volume measures reduce, up to ordinal equivalence, to the CES utility representation with geometrically normalized taste parameters. This normalization coincides with the one used in the CES price-index formula with changing tastes in Redding and Weinstein (2020).

The paper proceeds as follows. Section 2 presents the model and the contour axioms. Section 3 defines opportunity mappings and proves that opportunity-equivalent profiles, and in particular ray profiles, represent all contour-based well-being information on the relevant domain. Section 4 studies neutrality with respect to commodity labels and units, explains the arbitrariness of fixed rays and fixed prices, and establishes the impossibility results on unrestricted domains. Section 5 introduces finite-volume domains, states additive separability in ray profiles, and proves the volume representation theorem. Section 6 concludes.

2 Framework and contour axioms

2.1 Commodities, preferences, and situations

There are $L \geq 2$ divisible goods. The consumption set is $X = \mathbb{R}_+^L$ and its strictly positive part is denoted $X_{++} = \mathbb{R}_{++}^L$. For $x, y \in X$, write $x \geq y$ when $x_\ell \geq y_\ell$ for every good ℓ , write $x > y$ when $x \geq y$ and $x \neq y$, and write $x \gg y$ when $x_\ell > y_\ell$ for every ℓ .

A preference relation R is a complete and transitive ordering on X . Strict preference and indifference are denoted by P_R and $\sim_R$. The domain $\mathcal{R}$ consists of all orderings on X satisfying:

- (i) *continuity*: the graph of R is closed;
- (ii) *convexity*: every upper contour set is convex;
- (iii) *weak monotonicity*: if $x > y$, then xRy , and if $x \gg y$, then xP_Ry .

A *situation* is a pair $(x, R) \in X \times \mathcal{R}$. The bundle x is the bundle assigned to the individual, and R is the individual's preference relation. For every situation (x, R) define the indifference contour, upper contour set, and lower contour set through x by

$$I(x, R) = \{y \in X : y \sim_R x\},$$

$$U(x, R) = \{y \in X : yRx\}, \quad L(x, R) = \{y \in X : xRy\}.$$

Let

$$\mathcal{I} = \{I(x, R) : (x, R) \in X \times \mathcal{R}\}$$

be the collection of all indifference contours generated by the preference domain.

The following subset of contours will play a special role:

$$\mathcal{I}_{++} = \{I \in \mathcal{I} : \text{cl}(I \cap X_{++}) = I\},$$

where $\text{cl}(I \cap X_{++})$ denotes the closure of $I \cap X_{++}$. A contour in $\mathcal{I}_{++}$ is fully determined by its strictly positive part. For instance, the black indifference curve in Figure 2 belongs to $\mathcal{I}_{++}$, but the blue curve does not. Let

$$\mathcal{Z}_{++} = \{(x, R) \in X \times \mathcal{R} : I(x, R) \in \mathcal{I}_{++}\}.$$

The restriction to $\mathcal{Z}_{++}$ is innocuous if the primitive consumption set is X_{++} . In that case every contour satisfies $I \cap X_{++} = I$, and continuity implies that I is closed. It is also innocuous under global strong monotonicity on $X = \mathbb{R}_+^L$. Indeed, global strong monotonicity implies that the bottom contour is $I(0, R) = \{0\}$ for any preference. Conversely, every nonzero contour is the closure of its strictly positive part: any nonzero boundary point on such a contour can be approximated by strictly positive bundles on the same contour, using strong monotonicity and continuity. Thus, under global strong monotonicity, the only contour excluded from $\mathcal{I}_{++}$ is the bottom contour $\{0\}$.

Finally, a *well-being function* is a real-valued map

$$W : X \times \mathcal{R} \rightarrow \mathbb{R}.$$

It is meant to rank situations even when the underlying preferences differ. The goal of the paper is to characterize W from regularity and fairness axioms.

2.2 Basic axioms

The first condition is a minimal regularity requirement. It rules out jumps in the well-being number when the bundle changes slightly and preferences are held fixed.

Axiom 2.1 (Continuity). For every $R \in \mathcal{R}$, the map $x \mapsto W(x, R)$ is continuous on X .

The second condition is consumer sovereignty. Within a fixed preference relation, the well-being ranking must agree with the individual's own ranking of bundles.

Axiom 2.2 (Consumer sovereignty, or Pareto). For all $x, x' \in X$ and all $R \in \mathcal{R}$,

$$xRx' \implies W(x, R) \geq W(x', R), \quad xP_Rx' \implies W(x, R) > W(x', R).$$

The third condition is a contour dominance requirement taken from Fleurbaey and Maniquet (2017). It compares two situations such that their contours are unambiguously ordered - in the following sense. If no bundle can be simultaneously at least as good as x for R and at most as good as x' for R' , then the contour through (x, R) lies strictly above the contour through (x', R') . In that case, any preference-respecting interpersonal index must place (x, R) above (x', R') (see Figure 1).

Axiom 2.3 (Nested contour). For all $x, x' \in X$ and all $R, R' \in \mathcal{R}$,

$$U(x, R) \cap L(x', R') = \emptyset \quad \implies \quad W(x, R) > W(x', R').$$

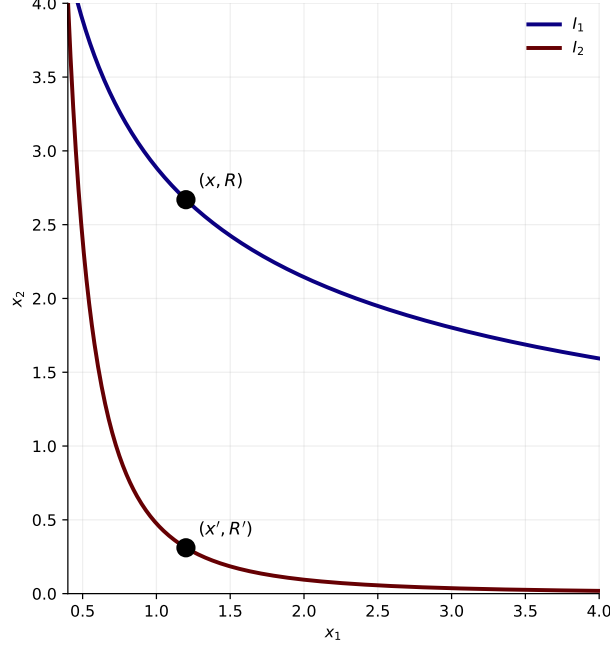


Figure 1: Under Nested contour: the well-being associated to (x, R) must be higher than the well-being associated to (x', R')

As shown in Fleurbaey and Maniquet (2017), Nested Contour has an equivalent informational formulation. If two preference relations generate the same indifference contour through the actual bundle, then they are assigned the same well-being level.

Axiom 2.4 (Unchanged contour independence). For all $x \in X$ and all $R, R' \in \mathcal{R}$,

$$I(x, R) = I(x, R') \quad \implies \quad W(x, R) = W(x, R').$$

Lemma 2.5 (Fleurbaey and Maniquet (2017)). *Under Continuity and Pareto, Nested Contour is equivalent to Unchanged Contour Independence.*

Lemma 2.5 transforms the interpersonal-comparison problem into a problem of comparing contours. Once Continuity and Pareto are imposed, a well-being function satisfies Nested Contour exactly when the indifference contour through the assigned bundle is a sufficient statistic for measured well-being. The next section shows that relevant information in a situation is then given by its *opportunity-equivalent profile*: the collection of indices of ordered opportunity sets whose best attainable element is just indifferent to the individual's actual bundle.

3 Opportunity-equivalent representations

3.1 Opportunity mappings

The equivalence approach evaluates a situation by asking which member of a reference family of opportunity sets is just sufficient to reach the individual's current contour. The reference family may consist of boxes, budget sets, or any other ordered collection of feasible sets.

Definition 3.1 (Opportunity mapping). An *opportunity mapping* is a correspondence

$$C : \mathbb{R}_+ \rightrightarrows X, \quad \lambda \mapsto C(\lambda),$$

satisfying the following conditions.

(C1) *Non-emptiness and compactness*. For every $\lambda \in \mathbb{R}_+$, $C(\lambda)$ is nonempty and compact.

(C2) *Continuity*. The correspondence C is continuous on nonempty compact subsets of X .

(C3) *Free disposal*. For every $\lambda \in \mathbb{R}_+$ and all $c, c' \in X$,

$$c \in C(\lambda), \ 0 \leq c' \leq c \implies c' \in C(\lambda).$$

(C4) *Strict expansion*. For all $\lambda < \lambda'$ and every $c \in C(\lambda)$, there exists $c' \in C(\lambda')$ such that $c' \gg c$.

(C5) *Exhaustiveness*. $C(0) = \{0\}$ and

$$\bigcup_{\lambda \geq 0} C(\lambda) = X.$$

Let $\mathcal{C}$ denote the class of all opportunity mappings.

Strict expansion is stronger than strict set inclusion: it says that every opportunity available at the lower index is strictly dominated by some opportunity available at the higher index. Therefore the best attainable preference level from $C(\lambda)$ rises strictly with λ for every monotone preference relation.

Fix $C \in \mathcal{C}$ and $R \in \mathcal{R}$. Let u be any continuous utility representation of R , and define

$$m_C^R(\lambda) = \max_{z \in C(\lambda)} u(z).$$

By compactness and continuity, the maximum exists. By the maximum theorem and strict expansion, m_C^R is continuous and strictly increasing. Since $C(0) = \{0\}$ and $\bigcup_{\lambda \geq 0} C(\lambda) = X$, every utility level $u(x)$ is reached by m_C^R at a unique opportunity index.

Definition 3.2 (Opportunity-equivalent well-being). For every $C \in \mathcal{C}$, the *opportunity-equivalent well-being function* generated by C is the map $W_C : X \times \mathcal{R} \rightarrow \mathbb{R}_+$ defined by

$$W_C(x, R) = \lambda \iff m_C^R(\lambda) = u(x),$$

where u is any continuous utility representation of R . Equivalently, $W_C(x, R) = \lambda$ if and only if there exists $z \in C(\lambda)$ such that $z \sim_R x$ and $z R y$ for every $y \in C(\lambda)$.

The number $W_C(x, R)$ is the size of the reference opportunity set whose best element is just indifferent to the individual's actual bundle. The construction is ordinal: replacing u by any increasing transformation leaves the equation $m_C^R(\lambda) = u(x)$ unchanged.

Lemma 3.3 (Basic properties of opportunity-equivalent well-being). *For every $C \in \mathcal{C}$, the function W_C satisfies Continuity, Pareto, and Unchanged Contour Independence.*

Two examples will be used repeatedly.

1. *Ray, or distance, functions.* For any $a \in X_{++}$, define

$$C_a(\lambda) = \{y \in X : y \leq \lambda a\}.$$

Because λa dominates every bundle in the box $C_a(\lambda)$, the associated opportunity-equivalent function is the unique scalar $D_a(x, R)$ satisfying

$$D_a(x, R)a \sim_R x.$$

Thus $D_a(x, R)$ is the distance one must travel along the positive ray spanned by a to reach the contour through x .

2. *Money-metric functions.* For any strictly positive price vector $\tilde{p} \in X_{++}$, define

$$C_{\tilde{p}}(\lambda) = \{y \in X : \tilde{p} \cdot y \leq \lambda\}.$$

The associated function, denoted $M_{\tilde{p}}$, is the expenditure level at prices $\tilde{p}$ whose indirect utility equals the utility of x .

3.2 Aggregating opportunity-equivalent profiles

Each opportunity mapping gives one possible interpersonal scale. Instead of selecting one such scale, one may retain the whole profile of opportunity-equivalent values.

Definition 3.4 (Opportunity-equivalent profile). For every situation (x, R) , define its *opportunity-equivalent profile*

$$V(x, R) : \mathcal{C} \rightarrow \mathbb{R}_+, \quad V(x, R)(C) = W_C(x, R).$$

For a domain $\mathcal{Y} \subseteq X \times \mathcal{R}$, write $V_{\mathcal{Y}}$ for the restriction of V to $\mathcal{Y}$.

An aggregator is a functional on the image of a profile map. The following monotonicity condition is the one needed to preserve consumer sovereignty.

Definition 3.5 (Admissible aggregator). A functional $\psi : \text{Im } V \rightarrow \mathbb{R}$ is *admissible* if, for all $v, v' \in \text{Im } V$,

- (i) if $v(C) \geq v'(C)$ for every $C \in \mathcal{C}$, then $\psi(v) \geq \psi(v')$;
- (ii) if $v(C) > v'(C)$ for every $C \in \mathcal{C}$, then $\psi(v) > \psi(v')$.

A well-being function W is an *aggregate opportunity-equivalent well-being function* if there exists an admissible aggregator ψ such that

$$W = \psi \circ V.$$

The aggregator may select one opportunity mapping, average several of them, take an infimum or supremum over a class, or apply a more general monotone functional. The important restriction is informational: only the opportunity-equivalent profile may be used.

Lemma 3.6 (Aggregation preserves the contour axioms). *Let $W = \psi \circ V$, where ψ is admissible. If W is continuous in x , then W satisfies Continuity, Pareto, and Unchanged Contour Independence.*

The converse representation holds on contours identified by their positive part.

Theorem 3.7 (Opportunity-equivalent representation). *Let $W : X \times \mathcal{R} \rightarrow \mathbb{R}$ satisfy Continuity, Pareto, and Unchanged Contour Independence. Then there exists an admissible aggregator $\psi : \text{Im } V_{\mathcal{Z}_{++}} \rightarrow \mathbb{R}$ such that*

$$W(x, R) = \psi(V(x, R)) \quad \text{for every } (x, R) \in \mathcal{Z}_{++}.$$

The theorem says that, once well-being depends only on the current contour, the complete family of opportunity-equivalent measures contains enough information to reconstruct the well-being comparison. There is no loss from passing through the opportunity-equivalent profile, except for boundary components that are invisible from the strictly positive orthant (see Figure 2). Though not excluded under weakly monotone preferences, we believe such boundary components to be pathological cases rather than relevant examples.

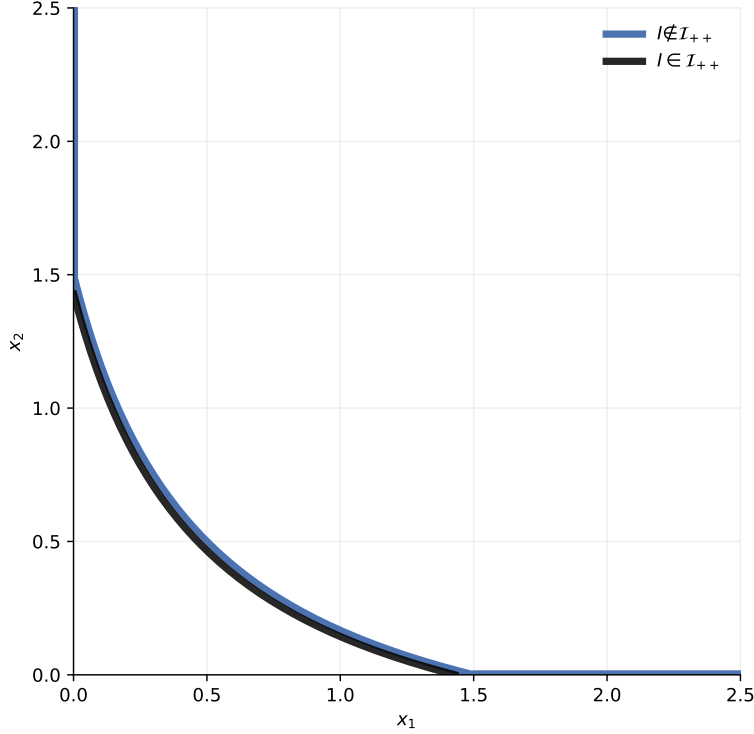


Figure 2: Two contours may coincide on X_{++} while differing on the boundary. Opportunity mappings cannot distinguish the boundary pieces.

In fact, the full class of opportunity mappings is more than is needed. Ray functions alone recover every contour in $\mathcal{I}_{++}$. Let $\widehat{X}_{++} \subset X_{++}$ be any normalized set of strictly positive rays, meaning that each set $\{ta : t > 0\}$ with $a \in X_{++}$ intersects $\widehat{X}_{++}$ exactly once. Examples include the simplex $\{a \in X_{++} : \sum_{\ell} a_{\ell} = 1\}$ and the product-normalized set $\{a \in X_{++} : \prod_{\ell} a_{\ell} = 1\}$.

Definition 3.8 (Ray profile). For every situation (x, R) , its *ray profile* is

$$\mathcal{D}(x, R) : \widehat{X}_{++} \rightarrow \mathbb{R}_{+}, \quad \mathcal{D}(x, R)(a) = D_a(x, R).$$

Corollary 3.9 (Ray representation). *Let $W : X \times \mathcal{R} \rightarrow \mathbb{R}$ satisfy Continuity, Pareto, and Nested Contour. Then there exists an aggregator ψ on $\text{Im } \mathcal{D}_{\mathcal{Z}_{++}}$ such that*

$$W|_{\mathcal{Z}_{++}} = \psi \circ \mathcal{D}|_{\mathcal{Z}_{++}}.$$

If one knows, for every positive ray, the unique point at which that ray meets the contour through x , then one knows the whole contour whenever the contour is the closure of its strictly positive part. The only information that can be missed lives entirely on the boundary of X , as in Figure 2. Note, however, that this ray representation is weaker than the opportunity-profile representation in one respect. The aggregator over ray profiles need not be strictly increasing in the pointwise ray distances. Two contours may be reached farther out on every strictly positive ray and nevertheless touch at the boundary. In such a case, the ray profile records a

strict improvement along every interior ray, but the closure of the contour still meets at a boundary contact. A contour-based well-being function may therefore assign the same value to the two situations, and the corresponding ray-profile aggregator must reproduce this equality, which is why the corollary only asserts the existence of an aggregator over ray profiles, not an admissible aggregator in the stronger sense used for the full opportunity-equivalent profile.

The same corollary could be formulated using the family of money-metric utilities instead of rays. For each strictly positive price vector, the money metric records the least expenditure required, at those prices, to reach the attained indifference contour. Varying the price vector over all strictly positive directions is therefore like looking at the contour through all possible budget hyperplanes. Since upper contour sets are closed and convex, this collection of supporting expenditure levels determines the upper contour set, and hence the indifference contour itself. Ray profiles and money-metric profiles can thus be seen as two dual ways of encoding the same contour information.

3.3 Money metrics as preference-sensitive ray minima

The ray representation clarifies the status of familiar well-being measures. A fixed ray function D_a selects one reference ray and evaluates everyone by the distance one must travel along that ray to reach the individual's indifference contour. This construction is transparent, but it can also be highly sensitive to the choice of the reference ray. The same contour can receive very different values depending on the direction chosen as reference. Figure 3 illustrates the point with two consumers who receive the same bundle but have different Leontief preferences. With the reference ray a , the individual whose indifference curve is I_2 appears much better off than the other consumer ($\lambda_2 > \lambda_1$). With the reference ray a' , the opposite comparison obtains ($\lambda'_1 > \lambda'_2$). The ranking is therefore driven not only by the two situations themselves, but also by an arbitrary external reference.

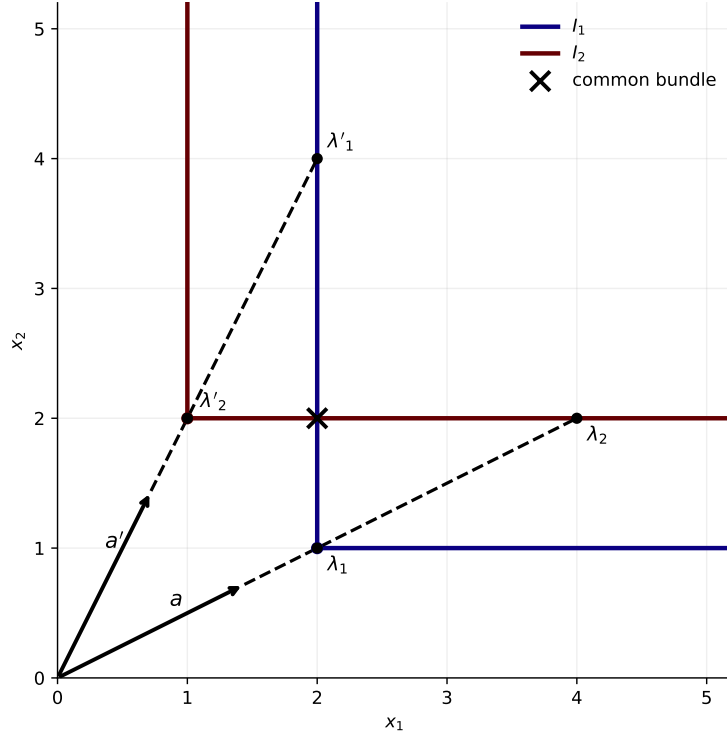


Figure 3: Ray utility can be very sensitive to the choice of reference ray.

A money metric seems, at first sight, to avoid this problem. It does not fix one ray in advance. Instead, it lets preferences select the cheapest point on the relevant contour, given a reference price vector. Fix $\tilde{p} \in X_{++}$ and define the price-normalized set of rays

$$\widehat{X}_{\tilde{p}} = \{a \in X_{++} : \tilde{p} \cdot a = 1\}.$$

Then, for every situation (x, R) ,

$$M_{\tilde{p}}(x, R) = \inf_{a \in \widehat{X}_{\tilde{p}}} D_a(x, R). \quad (1)$$

Indeed, if $\tilde{p} \cdot a = 1$, the bundle $D_a(x, R)a$ lies on the contour $I(x, R)$ and costs exactly $D_a(x, R)$ at prices $\tilde{p}$. Minimizing over price-normalized rays is therefore the same as finding the least expenditure needed to reach the contour.¹

This expression clarifies what makes money-metric utility attractive. It is a preference-sensitive ray measure of well-being: the reference bundle is not imposed externally, but is

¹Conversely, ray distances can be recovered from the full money-metric profile. Fix $a \in X_{++}$, and define the set of prices normalized by the cost of a :

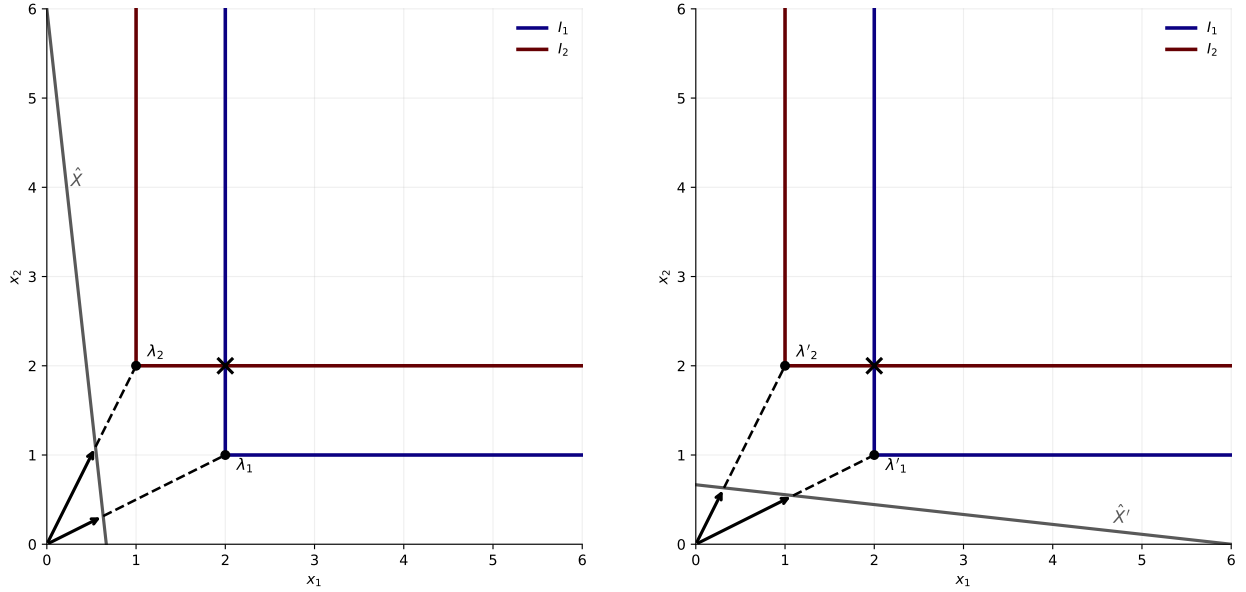
$$\widehat{P}_a = \{\tilde{p} \in X_{++} : \tilde{p} \cdot a = 1\}.$$

Then, for every situation (x, R) ,

$$D_a(x, R) = \sup_{\tilde{p} \in \widehat{P}_a} M_{\tilde{p}}(x, R). \quad (2)$$

selected as the cheapest way of reaching the individual's own indifference contour.² However, the word “cheapest” is meaningful only after all rays have been put on a common scale. The money metric does this by using the normalization $\tilde{p} \cdot a = 1$. This normalization determines which rays are large or small before the individual-specific minimization takes place.

Figure 4 illustrates the difficulty using the same Leontief example as Figure 3. One individual needs mostly good 1, the other needs mostly good 2, and both receive the same bundle. In this Leontief case, the ray selected by each individual is determined by her preferences and does not depend on the reference price vector. But the normalized length of that ray does depend on $\tilde{p}$. If the reference price makes good 2 cheap, the ray relevant for the second individual is normalized farther away from the origin, so the distance from the normalized ray to the contour is small. The second individual then appears worse off. If the reference price makes good 1 cheap, the reverse comparison obtains. The comparison is therefore driven by the normalization of rays, not by a substantive asymmetry between the two situations.



(a) Money metric with a cheap reference for good 2.

(b) Money metric with a cheap reference for good 1.

Figure 4: Money-metric utility can also be very sensitive to the normalization of reference rays.

One might try to restore symmetry by choosing the reference price vector $(1, 1)$ in the two-good case. But this only shifts the problem: the vector $(1, 1)$ is itself tied to a particular choice of units. Suppose, for example, that good 2 is milk. Should the symmetric reference price be $(1, 1)$ when milk is measured in liters, or when it is measured in half-liters? These two descriptions refer to the same physical good, but they lead to different coordinate

²This idea had already been formulated by Fleurbaey and Blanchet (2013) for non-income well-being dimensions.

normalizations and therefore to different reference prices. Hence, in a continuous commodity space, symmetry between goods cannot be implemented by simply choosing equal numerical prices unless the units themselves have independent normative significance. If no such privileged measuring scale is available, one possible response is to require the well-being ordering to be invariant to changes in commodity units. The rest of the paper studies whether one can construct well-being measures that are neutral in this stronger sense: both symmetric across goods and invariant to units of measurement.

4 Commodity Neutrality and Impossibility

4.1 Two neutrality axioms

The previous section discussed familiar contour-based well-being measures that depend on reference objects: a reference ray in the case of ray utilities, and a reference price vector in the case of money-metric utilities. The neutrality requirements introduced in this section are meant to rule out dependence on reference objects that have no independent ethical justification. They express the idea that, unless the model explicitly contains a normatively privileged coordinate system, neither the labels assigned to goods nor the relative numerical scales used to measure them should affect interpersonal well-being comparisons.

The first neutrality requirement is that goods should be treated symmetrically. The names or labels assigned to goods should not carry any ethical significance. Let π be a permutation of $\{1, \dots, L\}$. For $x \in X$, define

$$\pi x = (x_{\pi(1)}, \dots, x_{\pi(L)}).$$

For a preference relation R , define the permuted relation πR by

$$y (\pi R) z \iff \pi^{-1} y R \pi^{-1} z.$$

Thus $(\pi x, \pi R)$ is the same situation as (x, R) , except that the goods have been relabeled.

Axiom 4.1 (Symmetric treatment of goods). For every situation (x, R) and every permutation π ,

$$W(\pi x, \pi R) = W(x, R).$$

The axiom says that a well-being measure should not favor some goods merely because of the arbitrary order in which they are listed. For example, in a given system of units, a situation in which an individual receives one unit of good 1 and has the corresponding preference for good 1 should receive the same well-being number as the relabeled situation in which the individual receives one unit of good 2 and has the corresponding relabeled preference. Without this requirement, an index could favor “good-1 lovers” over “good-2 lovers” for purely notational reasons.

The second neutrality requirement concerns unit changes. Its interpretation is slightly more subtle. It asks whether the well-being ordering itself is independent of the relative

coordinate normalization of goods, without relying on an exogenous reference price, reference bundle, or any exogenously selected commodity scale.

Formally, let $\lambda = (\lambda_1, \dots, \lambda_L) \gg 0$ be a vector of positive coordinate rescalings. For $x \in X$, define

$$\lambda x = (\lambda_1 x_1, \dots, \lambda_L x_L).$$

For $R \in \mathcal{R}$, define the transformed relation λR by

$$y(\lambda R)z \iff \lambda^{-1}y R \lambda^{-1}z.$$

To motivate the formal statement, consider an initial coordinate system, denoted μ_0 , and let W_{μ_0} be the well-being function used to compare situations (x, R) whose commodity vectors are expressed in that coordinate system. This dependence on the coordinate system is usually suppressed, because the system of units is fixed once and for all. Now consider an alternative coordinate system μ_1 , obtained from the initial one by the relative rescaling λ . Unit Invariance says that the choice between these two coordinate systems should not affect the interpersonal ranking. Thus, for all situations (x, R) and (x', R') ,

$$W_{\mu_0}(x, R) \geq W_{\mu_0}(x', R') \iff W_{\mu_1}(x, R) \geq W_{\mu_1}(x', R').$$

But the physical situations that are represented by (x, R) and (x', R') in the coordinate system μ_1 are represented by $(\lambda x, \lambda R)$ and $(\lambda x', \lambda R')$ in the initial coordinate system μ_0 . Since what matters for well-being are the physical situations, consistency thus requires:

$$W_{\mu_1}(x, R) \geq W_{\mu_1}(x', R') \iff W_{\mu_0}(\lambda x, \lambda R) \geq W_{\mu_0}(\lambda x', \lambda R')$$

Finally, unit invariance can be written solely in the initial coordinate system as:

$$W_{\mu_0}(x, R) \geq W_{\mu_0}(x', R') \iff W_{\mu_0}(\lambda x, \lambda R) \geq W_{\mu_0}(\lambda x', \lambda R')$$

Suppressing the dependence on the fixed initial coordinate system yields the following axiom.

Axiom 4.2 (Unit Invariance). For all situations (x, R) and (x', R') , and every $\lambda \gg 0$,

$$W(x, R) \geq W(x', R') \iff W(\lambda x, \lambda R) \geq W(\lambda x', \lambda R').$$

The intuition is therefore that relative commodity scales should not be ethical primitives. If the only primitives are bundles and preferences, then a well-being comparison that depends on the particular normalization of the coordinate axes uses information that is not contained in the problem. The axiom is stronger than the claim that a measure is well-defined under changes of physical units. A money metric, for instance, is well defined under changes of units, as long as the reference price is made dependent on the units, but it is ruled out by Unit Invariance. Unless the reference price is itself given an independent normative role, the resulting interpersonal comparisons depend on a coordinate normalization that the present framework treats as arbitrary. This also indicates the normative limit of the axiom. If the environment does contain an ethically or economically meaningful reference scale - for example, a reference price vector representing social opportunity costs, a poverty-line bundle,

a needs threshold, or some other socially justified standard - then Unit Invariance may be too demanding. In that case, the reference scale is part of the normative problem rather than an arbitrary coordinate convention. The axiom is most defensible in the opposite case: when no such privileged normalization is available and the well-being ordering is required to be neutral with respect to all positive rescalings of the commodity coordinates.

Together, Symmetry and Unit Invariance impose a strong form of commodity neutrality. Symmetry says that commodity labels carry no ethical content. Unit Invariance says that the relative scales assigned to commodity coordinates carry no ethical content. The next section shows that these axioms - and more specifically Unit Invariance - are mathematically very demanding, leading to impossibility results on large preference domains.

4.2 Impossibility results

One may first try to extend the money-metric formula in Equation 1 in a way that satisfies symmetry and unit invariance.

Lemma 4.3. *Assume a well-being function is of the following ‘min-ray’ form*

$$W(x, R) = \inf_{a \in \widehat{X}_{++}} D_a(x, R). \quad (3)$$

where $\widehat{X}_{++}$ is a normalized set of strictly positive rays.³ Then W satisfies Symmetry and Unit Invariance if and only if, up to a positive multiplicative constant, $\widehat{X}_{++}$ is the product-normalized set:

$$\widehat{X}_{++} = \{a \in X_{++} : \prod_l a_l = 1\} \quad (4)$$

The product normalization is the only regular way to put rays on a common scale while respecting both neutrality requirements. It is symmetric across goods, and under a unit change λ it transforms by the common factor $\prod_l (\lambda_l)$. Hence the numerical index may change, but every interpersonal comparison is preserved.

Unfortunately, the product-normalized index collapses to zero for many contours that approach an axis. If a contour can be reached arbitrarily close to a boundary ray while maintaining $\prod_l a_l = 1$, the infimum in (3) is zero even though the individual may be strictly better off than another individual whose contour has the same property. This violates Pareto and leads to the following impossibility.

Proposition 4.4 (Failure of Pareto for neutral min-ray rules). *No min-ray index W on the full domain $X \times \mathcal{R}$ satisfies Pareto, Symmetry, and Unit Invariance.*

This is a first sign that neutrality and consumer sovereignty are difficult to combine on a rich preference domain. The next two results show that the difficulty is not specific to ‘min-ray’ rules.

³Formally, there exists a continuous, strictly increasing, homogeneous map $N : X_{++} \rightarrow \mathbb{R}_{++}$ such that

$$\widehat{X}_{++} = \{a \in X_{++} : N(a) = 1\}.$$

The money-metric normalization corresponds to $N(y) = p \cdot y$.

Proposition 4.5 (Impossibility on the full weakly monotone domain). *No well-being function on $X \times \mathcal{R}$ satisfies Pareto, Symmetry, and Unit Invariance.*

To form intuition, consider a framework with two goods and two consumers. Suppose one preference relation cares only about good 1 and another cares only about good 2. Symmetry equates the situation in which an individual receives one unit of good 1 and only values good 1 with the situation in which an individual receives one unit of good 2 and only values good 2. Unit Invariance says that this equality cannot depend on the units in which the two goods are measured. Any positive amount of good 1 can be renamed “one unit,” and the same is true for good 2. The equality therefore extends to all positive quantities. But then increasing the quantity of the good that the first individual actually values cannot raise her well-being, contradicting Pareto.

The preceding argument uses preferences that ignore some goods. The incompatibility survives even when all goods are strictly valuable in the interior.

Proposition 4.6 (Impossibility with strongly monotone preferences). *There exists no well-being function satisfying Pareto, Symmetry, and Unit Invariance on the domain of preferences that are strongly monotone on X_{++} .*

Thus a neutral and preference-respecting interpersonal index cannot exist on unrestricted domains. Any positive result must restrict preferences enough to rule out the rescaling arguments used in Propositions 4.5 and 4.6.

5 Volume Measures on Common-Tail Domains

The impossibility results show that the unrestricted preference domain is too large. We therefore turn to subdomains on which preferences are sufficiently well behaved near the axes and at infinity. Throughout this section, the preference subdomain under consideration will be assumed closed under permutations and positive unit changes (so that symmetry and unit invariance axioms are well defined). The notations $\mathcal{I}_{++}$ and $\mathcal{Z}_{++}$ are redefined relative to that subdomain. The goal is to characterize well-being functions on large subdomains where such functions exist. We first impose structure through a separability axiom introduced below.

5.1 Ray separability

For the rest of this section we use - without any loss of generality - the product-normalized ray set

$$\mathcal{S} = \left\{ a \in \mathbb{R}_{++}^L : \prod_{\ell=1}^L a_{\ell} = 1 \right\}.$$

Every strictly positive bundle $y \in X_{++}$ can be written uniquely as $y = ra$, with $r = (\prod_{\ell} y_{\ell})^{1/L}$ and $a \in \mathcal{S}$. Thus $\mathcal{S}$ is the set of directions once scale is measured by the geometric mean.

For any (x, R) , and any $a \in \mathcal{S}$, write $d_{x,R}(a) = D_a(x, R)$. The function $d_{x,R}$ is the radial representation of the indifference contour through x : on each normalized ray a , the contour is

reached at the bundle $d_{x,R}(a)a$. Ray separability says that the contribution of a ray depends on the distance at which the contour crosses that ray, and that total well-being is obtained by adding these contributions across rays. In finite dimensions, additive separability takes the form $\sum_i \phi_i(x_i)$. Here the index set of “coordinates” is the continuum of normalized rays, so the corresponding form is an integral over rays⁴.

Axiom 5.1 (Regular additive separability in the ray profile). Let $\mathcal{R}^* \subseteq \mathcal{R}$ be a preference domain. A well-being function $W : X \times \mathcal{R}^* \rightarrow \mathbb{R}$ is additively separable in the ray profile if the ordering induced by W on $X_{++} \times \mathcal{R}^*$ can be represented by a functional

$$F(d) = \int_S \phi_a(d(a)) \nu(da),$$

in the sense that, for all $(x, R), (x', R') \in X_{++} \times \mathcal{R}^*$,

$$W(x, R) \geq W(x', R') \iff F(d_{x,R}) \geq F(d_{x',R'}).$$

Here ν is a nonzero positive locally finite Borel measure with full support on S , the map $a \mapsto \phi_a(t)$ is measurable for every $t > 0$, and $\phi_a : \mathbb{R}_{++} \rightarrow \mathbb{R}$ is strictly increasing for ν -almost every a . Moreover, ϕ satisfies the regularity conditions stated in Appendix A.12.

The contour interpretation is direct. If two contours coincide on a set of rays, that common section contributes the same terms to both additive representations. Their comparison is therefore governed by the parts of the contours on which they differ, as in Figure 5.

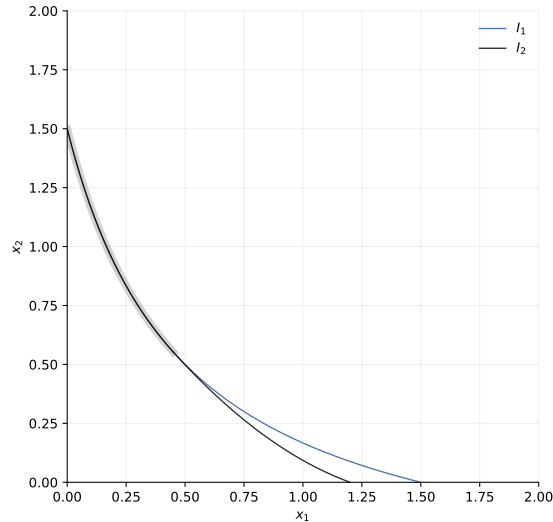


Figure 5: Under ray separability, the comparison of the two contours is determined by the sections on which the contours differ.

⁴In frameworks where the function has a finite number of arguments, this additive separability axiom is usually obtained as a result of an ordinal formulation of separability, through the Debreu-Gorman representation theorem. In frameworks with non-finite arguments, the generalization of this theorem was provided by Wakker and Zank 1999.

5.2 Weighted volumes of lower and upper contour sets

This subsection will characterize well-being functions that satisfy the neutrality and separability axioms introduced previously, on specific subdomains of preferences. It will involve the following object. For $\alpha \in \mathbb{R}$ and any measurable $A \subseteq X$, define the α -volume of A by

$$\text{Vol}_\alpha(A) = \int_{A \cap X_{++}} \prod_{\ell=1}^L y_\ell^{\alpha-1} dy.$$

- For $\bar{\alpha} \geq 0$, define the robust lower-volume domain

$$\mathcal{R}_{\bar{\alpha}}^+ = \{R \in \mathcal{R} : \text{Vol}_\alpha(L(x, R)) < \infty \text{ for every } x \in X_{++} \text{ and every } \alpha > \bar{\alpha}\}.$$

Intuitively, preferences that belong to robust lower-volume domains are those with indifference sets decaying asymptotically fast enough towards the axes, so that the lower contour-volumes stay finite. We call these substitute-type preferences. A necessary condition for a preference to be substitute-type is that every indifference set can be bounded from above by a ‘Cobb-Douglas boundary’, i.e. a set of product-normalized rays of the form $\prod_l x_l = c$ (as the solid curves in Figure 6a). The domains are nested: if $0 < \bar{\alpha}_1 \leq \bar{\alpha}_2$, then

$$\mathcal{R}_{\bar{\alpha}_1}^+ \subseteq \mathcal{R}_{\bar{\alpha}_2}^+.$$

- For $\bar{\alpha} \leq 0$, define the robust upper-volume domain

$$\mathcal{R}_{\bar{\alpha}}^- = \{R \in \mathcal{R} : \text{Vol}_\alpha(U(x, R)) < \infty \text{ for every } x \in X_{++} \text{ and every } \alpha < \bar{\alpha}\}.$$

For preferences to belong to robust upper-volume domains, the indifference sets must approach the boundaries with enough thickness (as in Figure 6b). We call these complement-type preferences. A necessary condition is that every indifference set is ‘above’ a set of the form $\prod_l x_l = c$. The domains are nested in the opposite direction: if $\bar{\alpha}_1 \leq \bar{\alpha}_2 < 0$, then

$$\mathcal{R}_{\bar{\alpha}_2}^- \subseteq \mathcal{R}_{\bar{\alpha}_1}^-.$$

The introduction of these domains is motivated by the following lemma:

Lemma 5.2 (No mixing of lower and upper-volume tails). *Let $\mathcal{R}^* \subseteq \mathcal{R}$ be a preference subdomain closed under permutations and positive unit changes. Suppose that there exist $R^+ \in \mathcal{R}^*$, $\bar{\alpha} \geq 0$, and $R^- \in \mathcal{R}^*$, $\bar{\alpha}' \leq 0$, such that*

$$R^+ \in \mathcal{R}_{\bar{\alpha}}^+ \quad \text{and} \quad R^- \in \mathcal{R}_{\bar{\alpha}'}^-.$$

Then no real-valued well-being function

$$W : X \times \mathcal{R}^* \rightarrow \mathbb{R}$$

satisfies Continuity, Pareto, Symmetry, Unit Invariance, and Additive Separability in the Ray Profile.

A positive result can thus only be achieved by confining preferences to robust lower volume domains or robust upper volume domains. The following representation theorem characterizes well-being functions defined on such domains.⁵

Theorem 5.3 (Volume representation). *The following statements hold.*

- (i) **Substitute-type preferences.** *Let $\bar{\alpha} \geq 0$ and let $W : X \times \mathcal{R}_{\bar{\alpha}}^+ \rightarrow \mathbb{R}$ be a real-valued well-being function satisfying Continuity, Pareto, Symmetry, Unit Invariance, and Additive Separability in the Ray Profile. Then there exists $\alpha > \bar{\alpha}$ such that*

$$W \sim \left(\text{Vol}_{\alpha}(L(x, R)) \right)^{1/\alpha}$$

on $X \times \mathcal{R}_{\bar{\alpha}}^+$, where $\sim$ denotes ordinal equivalence.

- (ii) **Complement-type preferences.** *Let $\bar{\alpha} \leq 0$ and let $W : X \times \mathcal{R}_{\bar{\alpha}}^- \rightarrow \mathbb{R}$ be a real-valued well-being function satisfying Continuity, Pareto, Symmetry, Unit Invariance, and Additive Separability in the Ray Profile. Then there exists $\alpha < \bar{\alpha}$ such that*

$$W \sim \begin{cases} \left(\text{Vol}_{\alpha}(U(x, R)) \right)^{1/\alpha}, & x \in X_{++}, \\ 0, & x \in \partial X. \end{cases}$$

on $X \times \mathcal{R}_{\bar{\alpha}}^-$.

The converse also holds: for every exponent in the indicated range, the function W satisfies all the listed axioms on its domain.

Lemma 5.2 and Theorem 5.3 have two main implications. First, the impossibility results in Propositions 4.5 and 4.6 can be avoided only by restricting attention to domains with a common tail geometry. More precisely, the admissible preferences must all fall on the same side of the Cobb-Douglas boundary: either their lower contour sets have finite weighted volume for the same positive orders, or their upper contour sets have finite weighted volume for the same negative orders. In the substitute-type case, indifference contours approach the axes fast enough for the lower contour sets to be measurable. In the complement-type case, they approach the axes slowly enough for the upper contour sets to be measurable. Thus a neutral separable index must aggregate either lower volumes or upper volumes. Such well-being functions can exist only on domains in which preferences exhibit a common substitutability pattern: either all admissible preferences are sufficiently substitute-like, so that all relevant lower volumes are finite, or all admissible preferences are sufficiently complement-like, so that all relevant upper volumes are finite. Mixed domains fall outside the scope of the representation because neither the lower-contour nor the upper-contour volume is finite for all admissible preferences.⁶

⁵Since additive separability is imposed directly on the ray profile, it strengthens the informational content of Nested Contour. Thus Nested Contour is not restated in the theorem; it is built into the ray-profile representation.

⁶One may question whether Symmetry is desirable on domains where all preferences agree that some goods play systematically different roles - for example, where some goods are always treated as substitutes and others as complements. In such domains, goods are not fully anonymous, and it may be normatively appropriate not to treat them as interchangeable.

Second, once this domain restriction and ray separability are imposed, the theorem leaves only one substantive degree of freedom: the exponent α . This exponent determines how distances along geometrically normalized rays are aggregated, as the next proposition makes explicit.

Proposition 5.4 (Radial formula for weighted volumes). *Let*

$$S = \left\{ a \in X_{++} : \prod_{\ell=1}^L a_{\ell} = 1 \right\}, \quad H = \left\{ z \in \mathbb{R}^L : \sum_{\ell=1}^L z_{\ell} = 0 \right\}.$$

For $z \in H$, write $a(z) = (e^{z_1}, \dots, e^{z_L}) \in S$, and let ds denote the measure on S induced by Lebesgue measure on H through the parametrization $z \mapsto a(z)$; that is, for every measurable function $h : S \rightarrow \mathbb{R}$,

$$\int_S h(a) ds(a) = \int_H h(a(z)) dz.$$

For every interior situation $(x, R) \in X_{++} \times \mathcal{R}^*$ with radial profile $d_{x,R} : S \rightarrow \mathbb{R}_{++}$,

$$\text{Vol}_{\alpha}(L(x, R)) \propto \int_S d_{x,R}(a)^{\alpha L} ds(a), \quad \alpha > 0,$$

whenever either side is finite. Similarly,

$$\text{Vol}_{\alpha}(U(x, R)) \propto \int_S d_{x,R}(a)^{\alpha L} ds(a), \quad \alpha < 0,$$

whenever either side is finite.

Writing $p = \alpha L$, the volume representatives are thus, up to irrelevant positive constants and monotone transformations, power-type integrals of the ray profile:

$$d \mapsto \left(\int_S d(a)^p ds(a) \right)^{1/p},$$

where ds is the Lebesgue measure in log-ratio coordinates.⁷ Given this way of counting rays, the parameter α determines how distances along those rays are aggregated. On the positive branch, large positive values of α emphasize long rays; on the negative branch, very negative values of α emphasize short rays. The normative content of this choice is better understood by asking which directions of the orthant correspond to long and short rays. These directions are not the same on the substitute and complement branches, as Figure 6 illustrates.

⁷Thus directions are counted uniformly in terms of relative changes in commodity proportions. In the two-good case, this means that the set of rays for which a_1/a_2 goes from 1 to 2 has the same ds -measure as the set of rays for which a_1/a_2 goes from 2 to 4: both correspond to the same change in $\log(a_1/a_2)$.

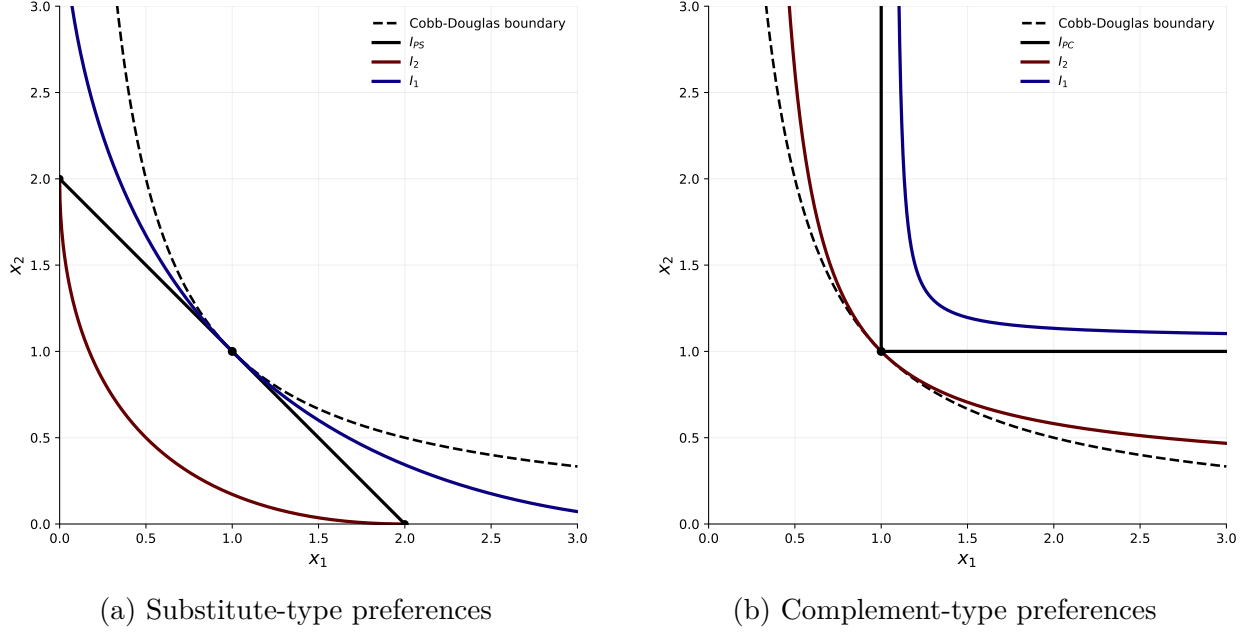


Figure 6: Illustration of the impact of α on well-being comparisons.

Each panel plots three indifference contours with different degrees of substitutability, together with the product-level curve that separates the lower- and upper-volume regimes. In panel [a](#), I_{PS} denotes the perfect-substitutes contour. In panel [b](#), I_{PC} denotes the perfect-complements contour.

Consider first substitute-type preferences, represented in Figure [6a](#). Such contours approach the axes faster than the product-level curves

$$\left\{ a \in \mathbb{R}_{++}^L : \prod_{\ell=1}^L a_{\ell} = c \right\},$$

shown by the dashed curve. As a result, the long rays (relatively to the product-normalized reference rays) have directions located away from the axes, in the interior of the orthant, whereas the short rays have directions close to the axes. For complement-type preferences, represented in Figure [6b](#), the opposite geometry obtains: the contours approach the axes more slowly than the product-level curves, so that the directions close to the axes correspond to long rays, while the interior directions correspond to short rays. Therefore, taking α toward $+\infty$ on the positive branch, and toward $-\infty$ on the negative branch, gives priority to directions in the interior of the orthant, away from the axes. This amounts to giving less importance to extreme bundles and more importance to balanced bundles. This is normatively defensible for convex preferences for which balanced bundles are more desirable than extreme bundles. In the limit $|\alpha| \rightarrow +\infty$, the comparisons illustrated in Figure [6](#) yield

$$I_1 \sim I_{\star},$$

where $I_{\star} = I_{PS}$ in the substitute panel and $I_{\star} = I_{PC}$ in the complement panel. In the substitute case, this amounts to treating perfect-substitutes preferences as advantageous relative to preferences with lower substitution possibilities: although I_1 lies above I_{PS} away

from the axes, the two situations are assigned the same well-being level. In the complement case, symmetrically, it leads to treating perfect-complements preferences as disadvantageous relative to preferences allowing more substitution.

By contrast, choosing a small absolute value of α , that is, taking α close to zero, gives more importance to directions close to the axes. This gives greater normative weight to extreme bundles. It treats preferences with greater substitution possibilities as disadvantageous - which may seem less defensible normatively. At the limit $\alpha \rightarrow 0$, the ranking becomes

$$I_\star \sim I_2.$$

On the positive branch, a natural benchmark between these two extreme positions is the ordinary-volume criterion, $\alpha = 1$. This is the case in which all bundles in the lower contour set are counted equally. In the configuration displayed in Figure 6a, it yields the intermediate ranking

$$I_1 \succ I_{PS} \succ I_2.$$

Thus $\alpha = 1$ corresponds to literal equal treatment of all alternatives in the lower contour set, while values below and above one tilt the comparison toward, respectively, directions close to the axes and directions in the interior of the orthant.

The choice of α therefore matters for comparing preferences with different degrees of substitutability between goods. A corollary is that the choice of α is innocuous for comparing preferences with identical elasticities of substitution, as discussed in the example below.

CES example. The volume representation takes a particularly simple form for CES preferences. Let $\rho = \frac{\sigma-1}{\sigma}$ and consider the CES preference relation R_θ represented by

$$u_\theta(x) = \left(\sum_{\ell=1}^L \theta_\ell x_\ell^\rho \right)^{1/\rho}, \quad \theta_\ell > 0,$$

with the usual limiting interpretation as $\rho \rightarrow 0$. Write

$$\bar{\theta} = \left(\prod_{\ell=1}^L \theta_\ell \right)^{1/L}$$

for the geometric mean of the taste parameters. Suppose that the elasticity of substitution is common across the individuals being compared, so that taste heterogeneity only occurs through the vectors θ . Then:

- if $\sigma > 1$, equivalently $\rho > 0$, preferences are of the substitute type. Lower contour sets have finite α -volume for every $\alpha > 0$. For every such α ,

$$(\text{Vol}_\alpha(L(x, R_\theta)))^{1/\alpha} \sim \bar{\theta}^{-1/\rho} u_\theta(x).$$

- if $\sigma < 1$, equivalently $\rho < 0$, preferences are of the complement type. Upper contour sets have finite α -volume for every $\alpha < 0$. For every such α ,

$$(\text{Vol}_\alpha(U(x, R_\theta)))^{1/\alpha} \sim \bar{\theta}^{-1/\rho} u_\theta(x).$$

- If $\sigma = 1$, equivalently $\rho = 0$, one obtains the Cobb-Douglas case. Cobb-Douglas contours lie exactly on the boundary between the two finite-volume regimes. Lower contour sets are too large near the axes to have finite positive α -volume, while upper contour sets are too large at infinity to have finite negative α -volume. Hence, for every α , neither the lower nor the upper contour sets have finite α -volume.

The dependence on α is therefore innocuous within the CES family, as long as $\sigma \neq 1$. The neutral volume criterion always ranks situations according to

$$\bar{\theta}^{-1/\rho} u_{\theta}(x),$$

regardless of α . Thus the axioms select the original CES utility representation, after a geometric normalization of the taste parameters. This is in fact exactly the geometric normalization that appears in the CES price-index formula of Redding and Weinstein (2020). The same example also nests Xu’s volume ranking of linear budget sets. Indeed, letting $\sigma \rightarrow +\infty$ yields the perfect-substitutes case, and linear budget sets are exactly the lower contour sets generated by perfect-substitutes preferences. The volume rule characterized by Xu (2004) is therefore recovered as the limiting CES case of the present framework.

6 Conclusion

This paper has studied how interpersonal well-being comparisons can be made when individuals have heterogeneous preferences. The first step of the paper was to characterize contour-based measures of well-being as functions of opportunity-equivalent indices. Each such index associates a situation with the member of a reference family of opportunity sets whose best attainable element is just indifferent to the individual’s actual bundle. Taken together, these indices provide a complete description of the attained contour on the relevant domain. In particular, it is enough to know the ray profile: the collection of distances at which normalized positive rays intersect the attained contour. This representation makes explicit the common structure behind familiar measures such as ray utilities and money-metric utilities. Both are ways of extracting information from an attained contour, but both rely on a reference object - a reference ray or a reference price vector. Unless such an object has independent normative content, the resulting interpersonal ranking depends on an arbitrary normalization.

The paper then asked whether this dependence on reference objects can be avoided. We imposed two neutrality requirements. Symmetry requires that commodity labels carry no ethical significance. Unit Invariance requires that the interpersonal well-being ordering be independent of the units used to measure commodities. These axioms are very demanding and require restrictions on preference domains to avoid impossibilities with Pareto. On finite-volume domains, and under additive separability in the ray profile, neutrality selects volume-type well-being measures. In substitute-type domains, the relevant object is the weighted volume of the lower contour set. In complement-type domains, the analogous object is the weighted volume of the upper contour set. These measures count, with a symmetric and unit-invariant density, the alternatives dominated by the individual’s actual bundle or the alternatives that dominate it.

The paper has focused only on the construction of individual well-being indices, and has not addressed how these indices should be aggregated across people in social evaluation. That is a separate normative question. The results obtained here are compatible with different social aggregators, from utilitarian to prioritarian or maximin criteria. What the paper provides is an input into such aggregation: a way of assigning individual well-being levels to heterogeneous-preference situations without choosing arbitrary interpersonal utility normalizations. This contribution, however, rests on the demanding Unit Invariance axiom, which is most appropriate when no reference scale has independent normative standing. There are many environments in which a reference price vector, a reference bundle, a poverty line, or a needs threshold may be part of the normative problem itself. In such cases, the use of a money metric or a reference-ray measure need not be arbitrary.

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A Proofs

A.1 Proof of Lemma 2.5

See Fleurbaey and Maniquet (2017).

A.2 Proof of Lemma 3.3

Proof. Let $C \in \mathcal{C}$ and $R \in \mathcal{R}$, and let u be a continuous utility representation of R . Compactness of $C(\lambda)$ and continuity of u imply that

$$m_C^R(\lambda) = \max_{z \in C(\lambda)} u(z)$$

is well-defined. The maximum theorem gives continuity of m_C^R . If $\lambda < \lambda'$, take a maximizer z of u on $C(\lambda)$. By strict expansion, there exists $z' \in C(\lambda')$ such that $z' \gg z$. Weak monotonicity gives $z' P_R z$, hence

$$m_C^R(\lambda') \geq u(z') > u(z) = m_C^R(\lambda).$$

Thus m_C^R is strictly increasing. Since $C(0) = \{0\}$ and the opportunity sets are exhaustive, every $u(x)$ lies in the range of m_C^R . Therefore $W_C(x, R)$ is well-defined and unique.

Because m_C^R is continuous and strictly increasing, its inverse is continuous on its range. Therefore

$$W_C(x, R) = (m_C^R)^{-1}(u(x))$$

is continuous in x .

If $x R x'$, then $u(x) \geq u(x')$, and the monotonicity of $(m_C^R)^{-1}$ gives $W_C(x, R) \geq W_C(x', R)$. If $x P_R x'$, the inequality is strict. Thus W_C satisfies Pareto.

Finally, the value $W_C(x, R)$ is determined by the lower side of the contour through x : $m_C^R(\lambda) \leq u(x)$ exactly when $C(\lambda) \subseteq L(x, R)$, and equality at the unique index means that $C(\lambda)$ touches $I(x, R)$ at a best point. $L(x, R)$ is determined by $I(x, R)$. Hence changing preferences away from the contour does not change $W_C(x, R)$. This is Unchanged Contour Independence. $\square$

A.3 Proof of Lemma 3.6

Proof. Continuity is assumed. Each coordinate W_C of the profile satisfies Pareto and UCI by Lemma 3.3. If $x R x'$, then $V(x, R)(C) \geq V(x', R)(C)$ for every $C \in \mathcal{C}$, and the admissibility of ψ gives $W(x, R) \geq W(x', R)$. If $x P_R x'$, then all coordinate inequalities are strict, and admissibility gives $W(x, R) > W(x', R)$. If $I(x, R) = I(x', R)$, every coordinate of the two profiles is the same, so the aggregate value is the same. Thus UCI holds. $\square$

A.4 Proof of Theorem 3.7 and Corollary 3.9

Proof. By UCI, there is a map $\widetilde{W} : \mathcal{I} \rightarrow \mathbb{R}$ such that

$$W(x, R) = \widetilde{W}(I(x, R)) \quad \text{for all } (x, R) \in X \times \mathcal{R}.$$

By Lemma 3.3, each opportunity-equivalent coordinate also factors through the contour, so there is a map $\widetilde{V} : \mathcal{I} \rightarrow \mathbb{R}_+^{\mathcal{C}}$ satisfying

$$V(x, R) = \widetilde{V}(I(x, R)).$$

The key point is injectivity on $\mathcal{I}_{++}$. Let $I_1, I_2 \in \mathcal{I}_{++}$ and suppose $\widetilde{V}(I_1) = \widetilde{V}(I_2)$. In particular, the profiles agree for all ray opportunity mappings. For every $a \in X_{++}$, the path $t \mapsto ta$ is strictly increasing in preferences, so each contour intersects the ray $\{ta : t \geq 0\}$ at a unique scalar. Equality of the ray coordinates implies that these scalars are the same for I_1 and I_2 . Hence

$$I_1 \cap X_{++} = I_2 \cap X_{++}.$$

Since both contours are closures of their strictly positive parts, $I_1 = I_2$.

Let φ be the restriction of $\widetilde{V}$ to $\mathcal{I}_{++}$. Since φ is injective, it has an inverse on its image. Define

$$\psi(v) = \widetilde{W}(\varphi^{-1}(v)) \quad \text{for } v \in \varphi(\mathcal{I}_{++}) = \text{Im } V_{Z_{++}}.$$

Then for every $(x, R) \in \mathcal{Z}_{++}$,

$$W(x, R) = \widetilde{W}(I(x, R)) = \psi(V(x, R)).$$

It remains to prove that ψ is admissible. Let $v, v' \in \text{Im } V_{Z_{++}}$, and let $I, I' \in \mathcal{I}^{++}$ be the unique contours such that

$$v = \widetilde{V}(I), \quad v' = \widetilde{V}(I'),$$

where $\widetilde{V}$ denotes the profile map written directly as a function of the attained contour. Write $L(I)$ and $U(I)$ for the lower and upper contour sets associated with I .

First suppose that

$$v(C) \geq v'(C) \quad \text{for every } C \in \mathcal{C}.$$

In particular this holds for all ray opportunity mappings. Hence, for every $a \in X_{++}$, the intersection of I with the ray $\{ta : t \geq 0\}$ is at least as far from the origin as the corresponding intersection of I' . Therefore

$$L(I') \cap X_{++} \subseteq L(I) \cap X_{++}.$$

Since $I' \in \mathcal{I}^{++}$, the set $L(I') \cap X_{++}$ is dense in $L(I')$. Indeed, if $y \in I'$, this follows from $\text{cl}(I' \cap X_{++}) = I'$; if $y \in L(I') \setminus I'$, then continuity implies that $y + \varepsilon \mathbf{1} \in L(I') \cap X_{++}$ for all sufficiently small $\varepsilon > 0$. Since $L(I)$ is closed, it follows that

$$L(I') \subseteq L(I).$$

We now show that this lower-contour inclusion implies the weak inequality between the corresponding W -values. Choose a situation (x, R) such that $I = I(x, R)$, and for $\varepsilon > 0$ set $x^\varepsilon = x + \varepsilon \mathbf{1}$. Since $x^\varepsilon P_R x$, we have

$$U(x^\varepsilon, R) \cap L(I) = \emptyset.$$

Because $L(I') \subseteq L(I)$, this implies

$$U(x^\varepsilon, R) \cap L(I') = \emptyset.$$

By Nested Contour, which follows from Continuity, Pareto and UCI by Lemma 2.5,

$$W(x^\varepsilon, R) > \widetilde{W}(I').$$

Letting $\varepsilon \downarrow 0$ and using Continuity of $W(\cdot, R)$, we obtain

$$\widetilde{W}(I) \geq \widetilde{W}(I').$$

Thus $\psi(v) \geq \psi(v')$.

It remains to prove the strict part. Suppose now that

$$v(C) > v'(C) \quad \text{for every } C \in \mathcal{C}.$$

By the weak part just proved, we again have $L(I') \subseteq L(I)$. We claim that

$$U(I) \cap L(I') = \emptyset.$$

Assume, to the contrary, that there exists $y \in U(I) \cap L(I')$. Since $L(I') \subseteq L(I)$, we have $y \in U(I) \cap L(I)$, hence $y \in I$.

We next show that $y \in I'$. If $y \in L(I') \setminus I'$, then y is strictly below the contour I' . By continuity, for all sufficiently small $\varepsilon > 0$,

$$y + \varepsilon \mathbf{1} \in L(I').$$

Since $L(I') \subseteq L(I)$, this gives $y + \varepsilon \mathbf{1} \in L(I)$. But $y \in I$ and $y + \varepsilon \mathbf{1} \gg y$, so monotonicity gives $y + \varepsilon \mathbf{1} \in U(I) \setminus L(I)$, a contradiction. Therefore $y \in I'$.

We now construct an opportunity mapping that detects this contact point. Since $I' \in \mathcal{I}^{++}$, there exists a sequence $(y^n)_{n \geq 1} \subset I' \cap X_{++}$ such that $y^n \rightarrow y$. Let

$$K = \{z \in X : \text{there exists } s \in \{y\} \cup \{y^n : n \geq 1\} \text{ such that } 0 \leq z \leq s\}.$$

Then K is nonempty, compact, and satisfies free disposal. Moreover $K \subseteq L(I')$, because every point of K is dominated by a point of I' . Since $L(I') \subseteq L(I)$, we also have $K \subseteq L(I)$, and because $y \in I \cap I'$, the set K touches both contours.

Define

$$C_K(\lambda) = \lambda K, \quad \lambda \geq 0.$$

Then $C_K \in \mathcal{C}$. Non-emptiness, compactness, continuity, free disposal, $C_K(0) = \{0\}$, and exhaustiveness follow immediately, because K contains at least one strictly positive point y^n .

Strict expansion also holds. Indeed, fix $0 \leq \lambda < \lambda'$ and $c = \lambda k \in C_K(\lambda)$. Put $\theta = \lambda/\lambda' < 1$. If k is dominated by some strictly positive y^n , then $y^n \gg \theta k$. If k is dominated only by y , choose n large enough that $y^n \gg \theta y$, which implies $y^n \gg \theta k$. Thus, in all cases, there exists $k' \in K$ such that $k' \gg \theta k$, and therefore $\lambda' k' \gg \lambda k = c$. Hence C_K satisfies strict expansion.

Since $K = C_K(1) \subseteq L(I')$ and K contains the point $y \in I'$, we have

$$\tilde{V}(I')(C_K) = 1.$$

Similarly, since $K \subseteq L(I)$ and $y \in I$,

$$\tilde{V}(I)(C_K) = 1.$$

Thus $v(C_K) = v'(C_K)$, contradicting the assumption that $v(C) > v'(C)$ for every $C \in \mathcal{C}$. Hence

$$U(I) \cap L(I') = \emptyset.$$

By Nested Contour,

$$\tilde{W}(I) > \tilde{W}(I').$$

Therefore $\psi(v) > \psi(v')$.

The two implications prove that ψ is admissible.

The proof for the ray profile is the same up to the injectivity step. The ray profile alone is injective on $\mathcal{I}^{++}$, because equality of ray distances on all strictly positive rays implies equality of the two contours on X_{++} , and contours in $\mathcal{I}^{++}$ are closures of their strictly positive parts. Hence the map D admits an inverse on its image, and the aggregator can be defined by

$$\psi(d) = \tilde{W}\left((D|_{\mathcal{I}^{++}})^{-1}(d)\right).$$

This proves $W|_{Z^{++}} = \psi \circ D|_{Z^{++}}$. □

A.5 A determinant-one implication of the neutrality axioms

For a unit change $\lambda \gg 0$, write

$$T_\lambda(x, R) = (\lambda x, \lambda R).$$

For a permutation π , write

$$T_\pi(x, R) = (\pi x, \pi R).$$

Lemma A.1. *If W satisfies Symmetry and Unit Invariance, then for every $t > 0$,*

$$W(tx_1, t^{-1}x_2, x_3, \dots, x_L; \mu R) = W(x, R),$$

where $\mu = (t, t^{-1}, 1, \dots, 1)$. Equivalently,

$$W(T_\mu z) = W(z) \quad \text{for every situation } z.$$

Proof. Let τ be the transposition exchanging goods 1 and 2. By Symmetry,

$$W(T_\tau T_\lambda z) = W(T_\lambda z)$$

for every z and every $\lambda \gg 0$. Unit Invariance preserves equality: applying $T_{\lambda^{-1}}$ to the two situations gives

$$W(T_{\lambda^{-1}} T_\tau T_\lambda z) = W(z).$$

Applying Symmetry once more gives

$$W(T_\tau T_{\lambda^{-1}} T_\tau T_\lambda z) = W(z).$$

For $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_L)$, the transformation on the left is the unit change associated with

$$\mu = \left(\frac{\lambda_1}{\lambda_2}, \frac{\lambda_2}{\lambda_1}, 1, \dots, 1 \right).$$

Choosing $\lambda_1/\lambda_2 = t$ proves the claim. □

A.6 Proof of Lemma 4.3

Proof. Let N be a regular ray normalization. For $b \in X_{++}$, let R^b be the Leontief preference represented by

$$u_b(y) = \min_\ell \frac{y_\ell}{b_\ell}.$$

The contour through b is

$$I(b, R^b) = \{y \in X : y \geq b \text{ and } y_\ell = b_\ell \text{ for at least one } \ell\}.$$

Since N is increasing, the infimum of N over this contour is attained at b . Hence

$$W_N(b, R^b) = N(b). \tag{A.1}$$

Assume W_N satisfies Unit Invariance. Applying Unit Invariance to two Leontief situations and using (A.1) gives, for all $b, c \in X_{++}$ and all $\lambda \gg 0$,

$$N(b) \geq N(c) \iff N(\lambda b) \geq N(\lambda c). \tag{A.2}$$

Thus $b \mapsto N(\lambda b)$ represents the same ordering as N . If $N(b) = N(c)$, (A.2) in both directions implies $N(\lambda b) = N(\lambda c)$. Therefore there exists a strictly increasing function $h_\lambda : \mathbb{R}_{++} \rightarrow \mathbb{R}_{++}$ such that

$$N(\lambda b) = h_\lambda(N(b)).$$

Homogeneity of N gives

$$h_\lambda(tN(b)) = N(\lambda tb) = tN(\lambda b) = th_\lambda(N(b)),$$

so $h_\lambda(s) = m(\lambda)s$ for some $m(\lambda) > 0$. Hence

$$N(\lambda b) = m(\lambda)N(b) \quad \text{for all } b, \lambda \in X_{++}. \tag{A.3}$$

Taking $b = \mathbf{1}$ shows $m(\lambda) = N(\lambda)/N(\mathbf{1})$, and therefore

$$N(\lambda b) = \frac{N(\lambda)N(b)}{N(\mathbf{1})}. \quad (\text{A.4})$$

Define $f : \mathbb{R}^L \rightarrow \mathbb{R}$ by

$$f(s) = \log N(e^{s_1}, \dots, e^{s_L}) - \log N(\mathbf{1}).$$

Equation (A.4) implies $f(s+t) = f(s) + f(t)$. Since N is continuous, f is continuous, and therefore $f(s) = \beta \cdot s$ for some $\beta \in \mathbb{R}^L$. Homogeneity of N implies $\sum_{\ell} \beta_{\ell} = 1$.

Now impose Symmetry. Applying Symmetry to the same Leontief test situations gives $N(\pi b) = N(b)$ for every permutation π . Hence the coefficients β_{ℓ} are all equal. Since they sum to one, $\beta_{\ell} = 1/L$ for every ℓ . Thus

$$N(b) = N(\mathbf{1}) \left(\prod_{\ell=1}^L b_{\ell} \right)^{1/L}.$$

Conversely, suppose $N(b) = \kappa(\prod_{\ell} b_{\ell})^{1/L}$. Then $N(\pi b) = N(b)$ for every permutation and

$$N(\lambda b) = \left(\prod_{\ell=1}^L \lambda_{\ell} \right)^{1/L} N(b) = \text{gm}(\lambda) N(b).$$

Using the contour-minimum formula (3), this implies

$$W_N(\pi x, \pi R) = W_N(x, R), \quad W_N(\lambda x, \lambda R) = \text{gm}(\lambda) W_N(x, R).$$

The first equality is Symmetry and the second equality preserves all interpersonal comparisons. Hence Unit Invariance holds. $\square$

A.7 Proof of Proposition 4.4

Proof. By lemma 4.3, any regular min-ray index satisfying Symmetry and Unit Invariance must use the product normalization up to scale. Consider the preference R^1 represented by the first coordinate, $u(y) = y_1$. For any $s > 0$ and any bundle x with $x_1 = s$, the contour is $\{y \in X : y_1 = s\}$. Along that contour one can take $y_2 \downarrow 0$ while keeping all other coordinates fixed and positive, so

$$\inf_{y \in I(x, R^1) \cap X_{++}} \left(\prod_{\ell=1}^L y_{\ell} \right)^{1/L} = 0.$$

Therefore $W_N(x, R^1) = 0$ for every $s > 0$. If $s' > s$, then the bundle with first coordinate s' is strictly preferred to the bundle with first coordinate s , but the well-being values are equal. Pareto is violated. $\square$

A.8 Proof of Proposition 4.5

Proof. Let R^1 be the preference relation represented by the first coordinate:

$$yR^1y' \iff y_1 \geq y'_1.$$

This relation belongs to $\mathcal{R}$: it is continuous, has convex upper contour sets, and satisfies weak monotonicity with strict improvements when all coordinates increase.

Let $x = (1, \dots, 1)$ and choose $t > 1$. Set $\mu = (t, t^{-1}, 1, \dots, 1)$. The preference relation is unchanged by this unit transformation:

$$y(\mu R^1)y' \iff \mu^{-1}y R^1 \mu^{-1}y' \iff \frac{y_1}{t} \geq \frac{y'_1}{t} \iff yR^1y'.$$

Thus $\mu R^1 = R^1$. Lemma A.1 gives

$$W(\mu x, R^1) = W(x, R^1).$$

But $\mu x = (t, t^{-1}, 1, \dots, 1)$ has a larger first coordinate than x , and R^1 only cares about the first coordinate. Hence $\mu x P_{R^1} x$. Pareto requires

$$W(\mu x, R^1) > W(x, R^1),$$

which contradicts the equality above. $\square$

A.9 Proof of Proposition 4.6

Proof. It is enough to consider Cobb-Douglas preferences with unequal exponents on two goods. Relabel goods so that $a_1 > a_2$, and let

$$u_a(y) = \prod_{\ell=1}^L y_\ell^{a_\ell}, \quad a_\ell > 0,$$

represent the preference relation R_a on X , with all boundary bundles assigned utility zero. This preference is continuous, has convex upper contour sets, and is strongly monotone on X_{++} .

Let $x = (1, \dots, 1)$ and choose $t > 1$. Set $\mu = (t, t^{-1}, 1, \dots, 1)$. For every $y \in X_{++}$,

$$u_a(\mu^{-1}y) = \left(\frac{y_1}{t}\right)^{a_1} (ty_2)^{a_2} \prod_{\ell=3}^L y_\ell^{a_\ell} = t^{a_2-a_1} u_a(y).$$

Multiplication by the positive constant $t^{a_2-a_1}$ does not change ordinal rankings, so $\mu R_a = R_a$. Lemma A.1 gives

$$W(\mu x, R_a) = W(x, R_a).$$

However,

$$u_a(\mu x) = t^{a_1-a_2} u_a(x) > u_a(x),$$

so $\mu x P_{R_a} x$. Pareto requires $W(\mu x, R_a) > W(x, R_a)$, a contradiction. $\square$

A.10 Proof of Proposition 5.4

Proof. Let

$$H = \left\{ z \in \mathbb{R}^L : \sum_{\ell=1}^L z_\ell = 0 \right\}.$$

For $z \in H$, write

$$a(z) = (e^{z_1}, \dots, e^{z_L}) \in \mathcal{S}.$$

Whenever we integrate uniformly over $\mathcal{S}$, we mean ordinary Lebesgue integration in the coordinates $z \in H$:

$$\int_{\mathcal{S}} h(a) ds(a) = \int_H h(a(z)) dz,$$

up to an irrelevant positive multiplicative constant.

Every $y \in X_{++}$ can be written uniquely as

$$y = ra, \quad r = \left(\prod_{\ell=1}^L y_\ell \right)^{1/L} > 0, \quad a \in \mathcal{S}.$$

Equivalently, $\log y = q\mathbf{1} + z$, with $q = \log r$ and $z \in H$. The Jacobian of the linear change of variables $(q, z) \mapsto q\mathbf{1} + z$ is a positive constant, and therefore

$$\prod_{\ell=1}^L y_\ell^{\alpha-1} dy = K_L e^{\alpha L q} dq dz = K_L r^{\alpha L - 1} dr ds(a).$$

Along a positive ray, the lower contour set is the segment $0 < r \leq d(a)$, while the upper contour set is the tail $r \geq d(a)$. Integrating first with respect to r gives the two formulas. $\square$

A.11 Proof of Lemma 5.2

Proof. Suppose, toward contradiction, that such a well-being function W exists. We work with the product-normalized ray set

$$S = \left\{ a \in \mathbb{R}_{++}^L : \prod_{\ell=1}^L a_\ell = 1 \right\}.$$

Equivalently, write

$$H = \left\{ z \in \mathbb{R}^L : \sum_{\ell=1}^L z_\ell = 0 \right\}, \quad a(z) = (e^{z_1}, \dots, e^{z_L}) \in S.$$

For an interior situation (x, R) , write its ray profile as

$$d_{x,R}(z) > 0, \quad d_{x,R}(z)a(z) \sim_R x.$$

By Additive Separability in the Ray Profile, the ordering induced by W on attainable ray profiles is represented by a real-valued additive functional

$$F(d) = \int_H \varphi_z(d(z)) \nu(dz),$$

where ν is a positive locally finite Borel measure with full support on H , and $\varphi_z : \mathbb{R}_{++} \rightarrow \mathbb{R}$ is strictly increasing for ν -almost every z . For each z , define the one-sided extended limits

$$\ell_z^0 = \lim_{t \downarrow 0} \varphi_z(t), \quad \ell_z^\infty = \lim_{t \uparrow \infty} \varphi_z(t),$$

and define the total variation of the z -coordinate by

$$V_z = \ell_z^\infty - \ell_z^0 \in [0, \infty].$$

Step 1: lower-volume profiles vanish at infinity, and upper-volume profiles diverge at infinity.

Let d be the ray profile of an interior situation (x, R^+) . Choose $\alpha > \bar{\alpha}$. Since $\bar{\alpha} \geq 0$, we have $\alpha > 0$. Because $R^+ \in \mathcal{R}_{\bar{\alpha}}^+$,

$$\text{Vol}_\alpha(L(x, R^+)) < \infty.$$

By the radial formula for weighted volumes, this implies

$$\int_H d(z)^{\alpha L} dz < \infty.$$

The ray profile is log-Lipschitz in z . Indeed, if

$$\delta = \|z - z'\|_\infty,$$

then

$$e^{-\delta} a(z) \leq a(z') \leq e^\delta a(z).$$

Since preferences are strictly increasing along every positive ray, this implies

$$e^{-\delta} d(z) \leq d(z') \leq e^\delta d(z). \tag{1}$$

If $d(z)$ did not converge to 0 as $\|z\| \rightarrow \infty$, there would exist $\varepsilon > 0$ and a sequence $z_n \rightarrow \infty$ such that

$$d(z_n) \geq \varepsilon.$$

Passing to a subsequence, we may assume that the balls $B(z_n, 1)$ are pairwise disjoint. By the log-Lipschitz estimate, d is bounded below by a positive constant on each $B(z_n, 1)$. Hence

$$\int_H d(z)^{\alpha L} dz = \infty,$$

a contradiction. Therefore

$$d(z) \rightarrow 0 \quad \text{as } \|z\| \rightarrow \infty. \tag{2}$$

Similarly, let g be the ray profile of an interior situation (y, R^-) . Choose $\beta < \bar{\alpha}'$. Since $\bar{\alpha}' \leq 0$, we have $\beta < 0$. Because $R^- \in \mathcal{R}_{\bar{\alpha}'}^-$,

$$\text{Vol}_\beta(U(y, R^-)) < \infty.$$

By the radial formula,

$$\int_H g(z)^{\beta L} dz < \infty.$$

Since $\beta L < 0$, this is integrability of a negative power of g . If $g(z)$ did not converge to $+\infty$ as $\|z\| \rightarrow \infty$, there would exist $M < \infty$ and a sequence $z_n \rightarrow \infty$ such that

$$g(z_n) \leq M.$$

Again taking disjoint balls and using the log-Lipschitz estimate, g would be bounded above by a fixed finite constant on each ball. Since $\beta L < 0$, $g^{\beta L}$ would then be bounded below by a positive constant on infinitely many disjoint balls, contradicting

$$\int_H g(z)^{\beta L} dz < \infty.$$

Thus

$$g(z) \rightarrow +\infty \quad \text{as } \|z\| \rightarrow \infty. \quad (3)$$

Step 2: determinant-one unit changes translate ray profiles without changing their F -value.

Symmetry and Unit Invariance imply invariance under every determinant-one unit change:

$$\prod_{\ell=1}^L \lambda_\ell = 1 \implies W(\lambda x, \lambda R) = W(x, R). \quad (4)$$

Indeed, Symmetry and Unit Invariance first imply invariance under a pairwise rescaling

$$(x_1, x_2, x_3, \dots, x_L) \mapsto (tx_1, t^{-1}x_2, x_3, \dots, x_L).$$

By conjugating this operation with permutations, the same conclusion holds for any pair of coordinates. Since every determinant-one unit change is a finite product of such pairwise rescalings, (4) follows.

Let $\lambda_\ell = e^{\tau_\ell}$ with $\tau \in H$. Then

$$\lambda a(z) = a(z + \tau).$$

Hence the ray profile of the transformed situation $(\lambda x, \lambda R)$ is

$$d_{\lambda x, \lambda R}(z) = d_{x, R}(z - \tau). \quad (5)$$

Since F represents the ordering induced by W , equality of W -values implies equality of F -values. Therefore, for every attainable ray profile d and every $\tau \in H$,

$$F(d(\cdot - \tau)) = F(d). \quad (6)$$

Step 3: the substitute-type preference forces infinite total variation near zero.

Choose $x_0, x_1 \in X_{++}$ with

$$x_1 \gg x_0.$$

Let

$$d_0 = d_{x_0, R^+}, \quad d_1 = d_{x_1, R^+}.$$

By Pareto,

$$W(x_1, R^+) > W(x_0, R^+),$$

and therefore

$$F(d_1) > F(d_0).$$

Moreover, by Step 1,

$$d_0(z) \rightarrow 0, \quad d_1(z) \rightarrow 0 \quad \text{as } \|z\| \rightarrow \infty.$$

In particular, d_1 is bounded. Also, since $x_1 P_{R^+} x_0$, we have

$$d_1(z) > d_0(z) \quad \text{for every } z \in H. \quad (7)$$

Fix $T > 0$. Suppose, toward contradiction, that

$$\int_H (\varphi_z(T) - \ell_z^0) \nu(dz) < \infty. \quad (8)$$

Choose $\eta > 0$ small enough that

$$\eta d_1(z) \leq T \quad \text{for every } z \in H.$$

The profiles ηd_0 and ηd_1 are attainable, because $\mathcal{R}^*$ is closed under common positive unit changes. By Pareto applied to the transformed preference,

$$\Delta := F(\eta d_1) - F(\eta d_0) > 0.$$

For every $\tau \in H$, using (6),

$$\Delta = \int_H [\varphi_z(\eta d_1(z - \tau)) - \varphi_z(\eta d_0(z - \tau))] \nu(dz). \quad (9)$$

The integrand is nonnegative by (7). Moreover, since

$$0 < \eta d_0(z - \tau) < \eta d_1(z - \tau) \leq T,$$

we have the domination

$$0 \leq \varphi_z(\eta d_1(z - \tau)) - \varphi_z(\eta d_0(z - \tau)) \leq \varphi_z(T) - \ell_z^0.$$

The right-hand side is integrable by (8).

Now take any sequence $\tau_n \in H$ with $\|\tau_n\| \rightarrow \infty$. For each fixed z ,

$$d_0(z - \tau_n) \rightarrow 0, \quad d_1(z - \tau_n) \rightarrow 0.$$

Therefore the integrand in (9) converges pointwise to 0. By dominated convergence, the right-hand side of (9) converges to 0, contradicting $\Delta > 0$. Hence (8) is false. Since $T > 0$ was arbitrary,

$$\int_H (\varphi_z(T) - \ell_z^0) \nu(dz) = \infty \quad \text{for every } T > 0. \quad (10)$$

Because

$$V_z = \ell_z^\infty - \ell_z^0 \geq \varphi_z(T) - \ell_z^0,$$

we obtain

$$\int_H V_z \nu(dz) = \infty. \quad (11)$$

Step 4: the complement-type preference makes real-valuedness impossible.

Let

$$d = d_0$$

be the substitute-type profile used above, and let

$$g = d_{y,R^-}$$

be the ray profile of any interior situation (y, R^-) . By Step 1,

$$d(z) \rightarrow 0 \quad \text{and} \quad g(z) \rightarrow +\infty \quad \text{as } \|z\| \rightarrow \infty.$$

Thus d is bounded above, and g is bounded away from zero. Hence we can choose $\eta > 0$ such that

$$\eta g(z) > d(z) \quad \text{for every } z \in H. \quad (12)$$

The profile ηg is attainable by a common positive unit change.

Since F is real-valued on attainable profiles,

$$C := F(\eta g) - F(d)$$

is finite. By determinant-one invariance, for every $\tau \in H$,

$$C = \int_H [\varphi_z(\eta g(z - \tau)) - \varphi_z(d(z - \tau))] \nu(dz). \quad (13)$$

The integrand is nonnegative by (12).

Take a sequence $\tau_n \in H$ with $\|\tau_n\| \rightarrow \infty$. For each fixed z ,

$$d(z - \tau_n) \rightarrow 0 \quad \text{and} \quad \eta g(z - \tau_n) \rightarrow +\infty.$$

Therefore

$$\varphi_z(\eta g(z - \tau_n)) - \varphi_z(d(z - \tau_n)) \rightarrow \ell_z^\infty - \ell_z^0 = V_z.$$

By Fatou's lemma and (13),

$$\int_H V_z \nu(dz) \leq \liminf_{n \rightarrow \infty} \int_H [\varphi_z(\eta g(z - \tau_n)) - \varphi_z(d(z - \tau_n))] \nu(dz) = C < \infty.$$

This contradicts (11), which says that

$$\int_H V_z \nu(dz) = \infty.$$

The contradiction proves that no such real-valued W exists. $\square$

A.12 Proof of Theorem 5.3

The following smoothness conditions are imposed in Axiom 5.1.

Assumption A.2 (Regularity conditions of the additive representation.). Let

$$H = \left\{ z \in \mathbb{R}^L : \sum_{\ell=1}^L z_\ell = 0 \right\}, \quad a(z) = (e^{z_1}, \dots, e^{z_L}) \in S.$$

If $d : S \rightarrow \mathbb{R}_{++}$ is an attainable interior ray profile, we write $d(z)$ for $d(a(z))$. Let ds denote Lebesgue measure on H , transported to S . The additive representation in Axiom 5.1 can be written as

$$F(d) = \int_H \phi_z(d(z)) \nu(dz),$$

where ν is a positive locally finite Borel measure with full support on H . We impose the following smoothness conditions on ϕ_z :

1.

$$\phi_z \in C^2(\mathbb{R}_{++}), \quad \phi'_z(t) > 0 \quad \text{for every } t > 0.$$

2. For every compact $K \subset H$ and every compact interval $J \subset \mathbb{R}_{++}$,

$$\int_K \sup_{t \in J} (|\phi'_z(t)| + |\phi''_z(t)|) \nu(dz) < \infty.$$

These technical assumptions will be used for the characterization of the weighted volume formulas. We start by proving the converse of the theorem, namely that the weighted volume functions satisfy all the axioms.

Step 0. Converse.

We prove that the volume formulas stated in the theorem satisfy the listed axioms. For every $\alpha \in \mathbb{R}$, write

$$\mu_\alpha(A) = \text{Vol}_\alpha(A) = \int_{A \cap X_{++}} \prod_{\ell=1}^L y_\ell^{\alpha-1} dy.$$

Thus μ_α is a Borel measure on X , absolutely continuous with respect to Lebesgue measure on X_{++} .

We first record two elementary facts that will be used repeatedly. Fix $R \in \mathcal{R}$, and let $u : X \rightarrow \mathbb{R}$ be a continuous utility representation of R . For each $t \in \mathbb{R}$, the level set

$$E_t = \{y \in X_{++} : u(y) = t\}$$

has μ_α -measure zero. Indeed, decompose $\mathbb{R}^L$ into affine lines parallel to $\mathbf{1} = (1, \dots, 1)$. On each such line, u is strictly increasing: if $y, y + s\mathbf{1} \in X_{++}$ and $s > 0$, then $y + s\mathbf{1} \gg y$, hence $y + s\mathbf{1} P_R y$. Therefore each level set E_t intersects each such line in at most one point. By Fubini, E_t has Lebesgue measure zero, and hence $\mu_\alpha(E_t) = 0$.

Consequently, whenever $t_n \rightarrow t$,

$$\mathbf{1}_{\{u \leq t_n\}} \rightarrow \mathbf{1}_{\{u \leq t\}} \quad \text{and} \quad \mathbf{1}_{\{u \geq t_n\}} \rightarrow \mathbf{1}_{\{u \geq t\}}$$

μ_α -almost everywhere.

We also note the transformation rule for μ_α . If $\lambda \gg 0$, then for every Borel set $A \subseteq X$,

$$\mu_\alpha(\lambda A) = \left(\prod_{\ell=1}^L \lambda_\ell^\alpha \right) \mu_\alpha(A). \quad (1)$$

This follows from the change of variables $y_\ell = \lambda_\ell z_\ell$. Similarly, for every permutation π ,

$$\mu_\alpha(\pi A) = \mu_\alpha(A). \quad (2)$$

Moreover,

$$L(\lambda x, \lambda R) = \lambda L(x, R), \quad U(\lambda x, \lambda R) = \lambda U(x, R), \quad (3)$$

and

$$L(\pi x, \pi R) = \pi L(x, R), \quad U(\pi x, \pi R) = \pi U(x, R). \quad (4)$$

Equations (1)-(4) also show that the domains $\mathcal{R}_\alpha^+$ and $\mathcal{R}_\alpha^-$ are closed under positive unit changes and permutations.

We now treat the two branches separately.

Lower-volume branch. Let $\bar{\alpha} \geq 0$, let $\alpha > \bar{\alpha}$, and define

$$W_+(x, R) = \left(\mu_\alpha(L(x, R)) \right)^{1/\alpha}$$

on $X \times \mathcal{R}_\alpha^+$. Since $\alpha > \bar{\alpha} \geq 0$, we have $\alpha > 0$.

First, W_+ is finite-valued. If $x \in X_{++}$, this is exactly the definition of $\mathcal{R}_\alpha^+$. If $x \in \partial X$, choose $b \in X_{++}$ such that $b \gg x$. Then $b P_R x$, so by transitivity

$$L(x, R) \subseteq L(b, R).$$

Since $b \in X_{++}$ and $R \in \mathcal{R}_\alpha^+$, $\mu_\alpha(L(b, R)) < \infty$. Hence

$$\mu_\alpha(L(x, R)) < \infty.$$

Next, W_+ satisfies Continuity. Fix $R \in \mathcal{R}_\alpha^+$, and let $x_n \rightarrow x$ in X . Choose $b \in X_{++}$ such that $b \gg x$ and, for all sufficiently large n , $b \gg x_n$. Then

$$L(x_n, R) \subseteq L(b, R)$$

for all sufficiently large n , and $\mu_\alpha(L(b, R)) < \infty$. Writing $t_n = u(x_n)$ and $t = u(x)$, we have $t_n \rightarrow t$, and

$$L(x_n, R) \cap X_{++} = \{y \in X_{++} : u(y) \leq t_n\}, \quad L(x, R) \cap X_{++} = \{y \in X_{++} : u(y) \leq t\}.$$

By the zero-level-mass observation above and dominated convergence,

$$\mu_\alpha(L(x_n, R)) \rightarrow \mu_\alpha(L(x, R)).$$

Since $v \mapsto v^{1/\alpha}$ is continuous on $\mathbb{R}_+$,

$$W_+(x_n, R) \rightarrow W_+(x, R).$$

We now verify Pareto. If xRx' , then

$$L(x', R) \subseteq L(x, R),$$

and therefore

$$W_+(x, R) \geq W_+(x', R).$$

If xP_Rx' , let $t = u(x)$ and $t' = u(x')$, so $t > t'$. Choose $c \in (t', t)$. Since X is convex and u is continuous, there exists $y^0 \in X$ with $u(y^0) = c$. For sufficiently small $\varepsilon > 0$,

$$y = y^0 + \varepsilon \mathbf{1} \in X_{++} \quad \text{and} \quad t' < u(y) < t.$$

By continuity of u , there is an open ball $B \subset X_{++}$ around y such that

$$t' < u(z) < t \quad \text{for every } z \in B.$$

Thus

$$B \subseteq L(x, R) \setminus L(x', R).$$

Since the density

$$y \mapsto \prod_{\ell=1}^L y_\ell^{\alpha-1}$$

is strictly positive on X_{++} , we have $\mu_\alpha(B) > 0$. Hence

$$\mu_\alpha(L(x, R)) > \mu_\alpha(L(x', R)),$$

and therefore

$$W_+(x, R) > W_+(x', R).$$

Symmetry follows from (2) and (4):

$$W_+(\pi x, \pi R) = \left(\mu_\alpha(\pi L(x, R)) \right)^{1/\alpha} = \left(\mu_\alpha(L(x, R)) \right)^{1/\alpha} = W_+(x, R).$$

Unit Invariance follows from (1) and (3). For every $\lambda \gg 0$,

$$\begin{aligned} W_+(\lambda x, \lambda R) &= \left(\mu_\alpha(\lambda L(x, R)) \right)^{1/\alpha} \\ &= \left[\left(\prod_{\ell=1}^L \lambda_\ell^\alpha \right) \mu_\alpha(L(x, R)) \right]^{1/\alpha} \\ &= \left(\prod_{\ell=1}^L \lambda_\ell \right) W_+(x, R). \end{aligned}$$

The multiplicative factor $\prod_\ell \lambda_\ell$ is positive and is the same for every situation. Hence, for all (x, R) and (x', R') ,

$$W_+(x, R) \geq W_+(x', R') \iff W_+(\lambda x, \lambda R) \geq W_+(\lambda x', \lambda R').$$

Finally, W_+ is additively separable in the ray profile. Let $p = \alpha L > 0$. For every interior situation $(x, R) \in X_{++} \times \mathcal{R}_\alpha^+$, let $d_{x,R}$ be the radial profile of the contour through x . By Proposition 5.4, there is a constant $C_{\alpha,L}^+ > 0$ such that

$$\mu_\alpha(L(x, R)) = C_{\alpha,L}^+ \int_H d_{x,R}(z)^p ds(z).$$

Therefore the ordering induced by W_+ is represented by

$$F_+(d) = \int_H d(z)^p ds(z).$$

This is an additive ray-profile representation with

$$\phi_z(t) = t^p \quad \text{and} \quad \nu = ds.$$

Since $p > 0$, $\phi'_z(t) = pt^{p-1} > 0$ for every $t > 0$. The function ϕ_z is C^2 , and the regularity conditions in the separability axiom hold because ds is locally finite and, on every compact interval $J \subset \mathbb{R}_{++}$, the functions $t \mapsto pt^{p-1}$ and $t \mapsto p(p-1)t^{p-2}$ are bounded. Thus Additive Separability in the Ray Profile holds.

Upper-volume branch. Let $\bar{\alpha} \leq 0$, let $\alpha < \bar{\alpha}$, and define

$$W_-(x, R) = \begin{cases} \left(\mu_\alpha(U(x, R)) \right)^{1/\alpha}, & x \in X_{++}, \\ 0, & x \in \partial X. \end{cases}$$

Since $\alpha < \bar{\alpha} \leq 0$, we have $\alpha < 0$.

For $x \in X_{++}$, $W_-(x, R)$ is finite-valued by the definition of $\mathcal{R}_{\bar{\alpha}}^-$. Moreover, it is strictly positive because $U(x, R)$ contains the upper orthant

$$Q(x) = \{y \in X_{++} : y \geq x\},$$

and

$$0 < \mu_\alpha(Q(x)) < \infty \quad (\alpha < 0, x \in X_{++}).$$

We first prove a useful implication of upper-volume finiteness. If $R \in \mathcal{R}_{\bar{\alpha}}^-$, then every interior bundle is strictly preferred to every boundary bundle, and all boundary bundles are mutually indifferent.

Indeed, let $b \in \partial X$ and $z \in X_{++}$. Suppose, toward a contradiction, that bRz . By weak monotonicity, every $y \in X_{++}$ with $y \geq b$ satisfies yRb , and hence yRz . Thus

$$\{y \in X_{++} : y \geq b\} \subseteq U(z, R).$$

Since b has at least one zero coordinate and $\alpha < 0$,

$$\mu_\alpha(\{y \in X_{++} : y \geq b\}) = \infty,$$

contradicting $\mu_\alpha(U(z, R)) < \infty$. Hence bRz is impossible. By completeness, $zP_R b$.

Now let $b, b' \in \partial X$. If $bP_R b'$, then, by continuity of a utility representation, $bP_R(b' + \varepsilon \mathbf{1})$ for all sufficiently small $\varepsilon > 0$. But $b' + \varepsilon \mathbf{1} \in X_{++}$, so the previous paragraph gives

$$b' + \varepsilon \mathbf{1} P_R b,$$

a contradiction. Hence no boundary bundle is strictly preferred to another. By completeness, all boundary bundles are indifferent.

We now prove Continuity of W_- . Fix $R \in \mathcal{R}_\alpha^-$. First let $x \in X_{++}$ and $x_n \rightarrow x$. For all sufficiently large n , $x_n \in X_{++}$. Choose $b \in X_{++}$ such that $x \gg b$ and, for all large n , $x_n \gg b$. Then

$$U(x_n, R) \subseteq U(b, R)$$

for all large n , and $\mu_\alpha(U(b, R)) < \infty$. Writing $t_n = u(x_n)$ and $t = u(x)$, dominated convergence and the zero-level-mass observation give

$$\mu_\alpha(U(x_n, R)) \rightarrow \mu_\alpha(U(x, R)).$$

Since $v \mapsto v^{1/\alpha}$ is continuous on $\mathbb{R}_{++}$,

$$W_-(x_n, R) \rightarrow W_-(x, R).$$

Now let $x \in \partial X$ and $x_n \rightarrow x$. If $x_n \in \partial X$, then $W_-(x_n, R) = 0$. If $x_n \in X_{++}$, then

$$Q(x_n) = \{y \in X_{++} : y \geq x_n\} \subseteq U(x_n, R),$$

so

$$\mu_\alpha(U(x_n, R)) \geq \mu_\alpha(Q(x_n)).$$

For $x_n \in X_{++}$,

$$\mu_\alpha(Q(x_n)) = \prod_{\ell=1}^L \int_{x_{n\ell}}^{\infty} y_\ell^{\alpha-1} dy_\ell = \left(-\frac{1}{\alpha}\right)^L \prod_{\ell=1}^L x_{n\ell}^\alpha.$$

Since $x \in \partial X$, at least one coordinate of x_n converges to zero. Because $\alpha < 0$,

$$\mu_\alpha(Q(x_n)) \rightarrow \infty.$$

Thus, along the interior subsequence,

$$0 \leq W_-(x_n, R) = \left(\mu_\alpha(U(x_n, R))\right)^{1/\alpha} \leq \left(\mu_\alpha(Q(x_n))\right)^{1/\alpha} \rightarrow 0,$$

because $1/\alpha < 0$. Hence $W_-(x_n, R) \rightarrow 0 = W_-(x, R)$. Continuity holds on all of X .

We verify Pareto. Let xRx' .

If $x, x' \in X_{++}$, then

$$U(x, R) \subseteq U(x', R),$$

so

$$\mu_\alpha(U(x, R)) \leq \mu_\alpha(U(x', R)).$$

Since $1/\alpha < 0$, the map $v \mapsto v^{1/\alpha}$ is strictly decreasing, and therefore

$$W_-(x, R) \geq W_-(x', R).$$

If, moreover, $xP_R x'$, then $u(x) > u(x')$. As in the lower-volume case, there is an open ball $B \subset X_{++}$ such that

$$u(x') < u(y) < u(x) \quad \text{for every } y \in B.$$

Hence

$$B \subseteq U(x', R) \setminus U(x, R),$$

so

$$\mu_\alpha(U(x', R)) > \mu_\alpha(U(x, R)).$$

Because $1/\alpha < 0$,

$$W_-(x, R) > W_-(x', R).$$

If $x \in X_{++}$ and $x' \in \partial X$, then the boundary claim above gives $xP_R x'$, and

$$W_-(x, R) > 0 = W_-(x', R).$$

If $x, x' \in \partial X$, then the boundary claim gives $x \sim_R x'$, and

$$W_-(x, R) = W_-(x', R) = 0.$$

Finally, the case $x \in \partial X$ and $x' \in X_{++}$ cannot occur under the hypothesis xRx' , because every interior bundle is strictly preferred to every boundary bundle. Thus Pareto holds.

Symmetry follows exactly as in the lower-volume branch. If $x \in X_{++}$, then

$$W_-(\pi x, \pi R) = \left(\mu_\alpha(\pi U(x, R)) \right)^{1/\alpha} = W_-(x, R),$$

and if $x \in \partial X$, then also $\pi x \in \partial X$, so both sides are zero.

Unit Invariance also follows from the same transformation rule. If $x \in X_{++}$, then

$$\begin{aligned} W_-(\lambda x, \lambda R) &= \left(\mu_\alpha(\lambda U(x, R)) \right)^{1/\alpha} \\ &= \left[\left(\prod_{\ell=1}^L \lambda_\ell^\alpha \right) \mu_\alpha(U(x, R)) \right]^{1/\alpha} \\ &= \left(\prod_{\ell=1}^L \lambda_\ell \right) W_-(x, R). \end{aligned}$$

If $x \in \partial X$, then $\lambda x \in \partial X$, and the same identity holds because both sides are zero. Therefore, for every $\lambda \gg 0$,

$$W_-(\lambda x, \lambda R) = \left(\prod_{\ell=1}^L \lambda_\ell \right) W_-(x, R)$$

for every situation (x, R) . Since the factor is positive and common to all situations, Unit Invariance holds.

It remains to verify Additive Separability in the Ray Profile. Let $p = \alpha L < 0$. By the boundary claim, boundary contours in the upper-volume branch contain no strictly positive point: every boundary bundle is indifferent to every other boundary bundle, and every interior bundle is strictly preferred to every boundary bundle. Axiom 5.1 applies only to interior situations, so it remains to check the additive representation on $X_{++} \times \mathcal{R}_{\bar{\alpha}}^-$.

For such x , Proposition 5.4 gives a constant $C_{\alpha,L}^- > 0$ such that

$$\mu_{\alpha}(U(x, R)) = C_{\alpha,L}^- \int_H d_{x,R}(z)^p ds(z).$$

Since $1/\alpha < 0$, the ordering induced by W_- is represented by

$$F_-(d) = - \int_H d(z)^p ds(z).$$

This is additively separable with

$$\phi_z(t) = -t^p \quad \text{and} \quad \nu = ds.$$

Because $p < 0$,

$$\phi'_z(t) = -pt^{p-1} > 0 \quad \text{for every } t > 0.$$

Again $\phi_z \in C^2(\mathbb{R}_{++})$, and the regularity conditions in the separability axiom hold on compact subsets because ds is locally finite and the derivatives of $t \mapsto -t^p$ are bounded on compact intervals $J \subset \mathbb{R}_{++}$.

Therefore W_- satisfies Continuity, Pareto, Symmetry, Unit Invariance, and Additive Separability in the Ray Profile. This proves the converse for the upper-volume branch, and hence completes the proof of the theorem.

We now prove the direct implication.

Proof. Throughout the proof, ray profiles are written in the product-normalized coordinates

$$H = \left\{ z \in \mathbb{R}^L : \sum_{\ell=1}^L z_{\ell} = 0 \right\}, \quad a(z) = (e^{z_1}, \dots, e^{z_L}) \in \mathcal{S}.$$

On $X_{++} \times \mathcal{R}_{\bar{\alpha}}^{\pm}$, Additive Separability in the Ray Profile gives an additive representative

$$F(d) = \int_H \phi_z(d(z)) \nu(dz),$$

where ν is a positive locally finite Borel measure with full support on H , and where the regularity assumptions stated in the axiom hold.

Step 1. Local richness of the ray-profile domains.

Lemma A.3 (Local profile richness). *For each robust volume branch, there exists a smooth profile $b : H \rightarrow \mathbb{R}_{++}$ such that, for every compact $K \subset H$, there exists $\varepsilon > 0$ with the following property: if $\eta \in C_c^2(K)$ and $\|\eta\|_{C^2} < \varepsilon$, then*

$$d_\eta(z) = b(z)e^{\eta(z)}$$

is the ray profile of some admissible situation in the relevant branch.

Moreover, the same property holds around every translated and rescaled profile

$$d^{c,\tau}(z) = cb(z - \tau), \quad c > 0, \tau \in H.$$

Proof. For the lower-volume branch, choose $0 < \rho < 1$. For the upper-volume branch, choose $\rho < 0$. In both cases consider the CES utility

$$u_\rho(y) = \left(\sum_{\ell=1}^L y_\ell^\rho \right)^{1/\rho}.$$

It is continuous, strictly increasing on X_{++} , homogeneous of degree one, and concave. Writing $y = ra(z)$, homogeneity gives

$$u_\rho(ra(z)) = rh_\rho(z), \quad h_\rho(z) = \left(\sum_{\ell=1}^L e^{\rho z_\ell} \right)^{1/\rho}.$$

The unit indifference contour has ray profile

$$b(z) = h_\rho(z)^{-1} = \left(\sum_{\ell=1}^L e^{\rho z_\ell} \right)^{-1/\rho}.$$

Let $K \subset H$ be compact and let $\eta \in C_c^2(K)$. Define

$$d_\eta(z) = b(z)e^{\eta(z)}.$$

Equivalently, define

$$h_\eta(z) = d_\eta(z)^{-1} = h_\rho(z)e^{-\eta(z)}$$

and the homogeneous function

$$u_\eta(ra(z)) = rh_\eta(z).$$

The perturbation is compactly supported in the angular variable. On compact sets of directions, strict monotonicity and strict convexity of the upper contour of the CES utility are open C^2 -properties. Hence, for $\|\eta\|_{C^2}$ small enough, the set

$$C_\eta = \{ra(z) : r \geq d_\eta(z)\}$$

is still a closed, convex, comprehensive upper set with smooth boundary. Thus u_η represents a continuous, convex, strictly monotone preference relation on X_{++} .

Because η is compactly supported in H , the perturbed profile has the same tails as the CES profile. Hence it remains in the same robust volume branch. Common rescalings multiply ray profiles by a positive constant, and determinant-one unit changes translate z . Since the robust volume branches are closed under these operations, the same local richness property holds around every profile $cb(\cdot - \tau)$. $\square$

Step 2. Local uniqueness of additive representations.

Lemma A.4. *Let $I, J \subset \mathbb{R}$ be nonempty open intervals. Let $P : I \rightarrow \mathbb{R}$, $Q : J \rightarrow \mathbb{R}$, and $\Theta : I + J \rightarrow \mathbb{R}$ satisfy*

$$P(x) + Q(y) = \Theta(x + y) \quad \text{for all } x \in I, y \in J. \quad (5)$$

Assume P and Q are continuous and Θ is increasing. Then there exist constants $A, B_P, B_Q \in \mathbb{R}$ such that

$$P(x) = Ax + B_P, \quad Q(y) = Ay + B_Q, \quad \Theta(r) = Ar + B_P + B_Q$$

for all $x \in I$, $y \in J$, and $r \in I + J$. If Θ is strictly increasing, then $A > 0$.

Proof. Fix $x_0 \in I$ and $y_0 \in J$, and put $r_0 = x_0 + y_0$. For u, v sufficiently small, with $x_0 + u \in I$, $y_0 + v \in J$, and $u + v$ still sufficiently small, Equation 5 gives

$$\begin{aligned} \Theta(r_0 + u + v) &= P(x_0 + u) + Q(y_0 + v), \\ \Theta(r_0 + u) &= P(x_0 + u) + Q(y_0), \\ \Theta(r_0 + v) &= P(x_0) + Q(y_0 + v), \\ \Theta(r_0) &= P(x_0) + Q(y_0). \end{aligned}$$

Subtracting the second and third equations from the first and adding the fourth yields

$$\Theta(r_0 + u + v) - \Theta(r_0 + u) - \Theta(r_0 + v) + \Theta(r_0) = 0.$$

Thus the function

$$A_{r_0}(w) = \Theta(r_0 + w) - \Theta(r_0)$$

satisfies the Cauchy equation

$$A_{r_0}(u + v) = A_{r_0}(u) + A_{r_0}(v)$$

locally around 0. Since Θ is increasing, A_{r_0} is increasing. The monotone local solutions of Cauchy's equation are linear, so there is a number $a(r_0)$ such that

$$\Theta(r_0 + w) = \Theta(r_0) + a(r_0)w$$

for all w in a neighborhood of 0. Hence Θ is locally affine on the interval $I + J$. Since $I + J$ is connected, the local affine representations agree on overlaps. Therefore there are constants $A, B \in \mathbb{R}$ such that

$$\Theta(r) = Ar + B \quad \text{for all } r \in I + J.$$

Substituting this expression into 5 gives

$$P(x) + Q(y) = A(x + y) + B.$$

Fixing $y = y_0$ shows that $P(x) = Ax + B_P$, and fixing $x = x_0$ shows that $Q(y) = Ay + B_Q$. If Θ is strictly increasing, then necessarily $A > 0$. $\square$

Lemma A.5 (Local additive uniqueness). *Let $O \Subset H$ be a nonempty relatively compact open set, and let $d^0 : H \rightarrow \mathbb{R}_{++}$ be a smooth profile. For $\varepsilon > 0$, define the local profile chart*

$$\mathcal{U}_\varepsilon(O) = \left\{ d_u = d^0 e^u : u \in C_c^2(O), \|u\|_{C^2} < \varepsilon \right\}.$$

Let

$$F(d) = \int_H f_z(d(z)) \mu(dz), \quad G(d) = \int_H g_z(d(z)) \eta(dz)$$

be two regular additive functionals, finite on $\mathcal{U}_\varepsilon(O)$, where μ and η are positive locally finite Borel measures with full support on H . Assume that, for μ -almost every z , $f_z \in C^1(\mathbb{R}_{++})$ and $f'_z(t) > 0$ for every $t > 0$, and that, for η -almost every z , $g_z \in C^1(\mathbb{R}_{++})$ and $g'_z(t) > 0$ for every $t > 0$. Assume also the usual local domination condition: for every compact interval $J \Subset \mathbb{R}_{++}$,

$$\int_O \sup_{t \in J} |f'_z(t)| \mu(dz) < \infty, \quad \int_O \sup_{t \in J} |g'_z(t)| \eta(dz) < \infty.$$

Suppose that F and G represent the same ordering on $\mathcal{U}_\varepsilon(O)$: for all $d_u, d_v \in \mathcal{U}_\varepsilon(O)$,

$$F(d_u) \geq F(d_v) \iff G(d_u) \geq G(d_v).$$

Then, after possibly replacing ε by a smaller positive number, there exist constants $A > 0$ and $B \in \mathbb{R}$ such that

$$G(d_u) = AF(d_u) + B \quad \text{for every } d_u \in \mathcal{U}_\varepsilon(O).$$

Consequently, for every $d_u \in \mathcal{U}_\varepsilon(O)$,

$$g'_z(d_u(z)) \eta(dz) = Af'_z(d_u(z)) \mu(dz)$$

as locally finite Borel measures on O .

Proof. Since F and G represent the same ordering, equality of F -values implies equality of G -values. Indeed, if $F(d_u) = F(d_v)$, then both $F(d_u) \geq F(d_v)$ and $F(d_v) \geq F(d_u)$, hence both $G(d_u) \geq G(d_v)$ and $G(d_v) \geq G(d_u)$, so $G(d_u) = G(d_v)$.

Therefore there is a well-defined strictly increasing function h on the local range

$$\mathcal{I} = F(\mathcal{U}_\varepsilon(O))$$

such that

$$G(d_u) = h(F(d_u)) \quad \text{for every } d_u \in \mathcal{U}_\varepsilon(O).$$

The set $\mathcal{U}_\varepsilon(O)$ is convex, hence connected, and F is continuous on it by the domination assumptions. Thus $\mathcal{I}$ is an interval.

We now prove that h is locally affine on $\mathcal{I}$. Fix a profile $d_{\bar{u}} \in \mathcal{U}_\varepsilon(O)$. Choose two nonzero nonnegative functions $v, w \in C_c^2(O)$ with disjoint supports. For s, t sufficiently close to zero, define

$$d_{s,t}(z) = d_{\bar{u}}(z) e^{sv(z)+tw(z)}.$$

Because the supports of v and w are disjoint, additivity gives

$$F(d_{s,t}) = F(d_{\bar{u}}) + A_1(s) + A_2(t),$$

where

$$\begin{aligned} A_1(s) &= \int_{\text{supp } v} \left[f_z(d_{\bar{u}}(z)e^{sv(z)}) - f_z(d_{\bar{u}}(z)) \right] \mu(dz), \\ A_2(t) &= \int_{\text{supp } w} \left[f_z(d_{\bar{u}}(z)e^{tw(z)}) - f_z(d_{\bar{u}}(z)) \right] \mu(dz). \end{aligned}$$

Similarly,

$$G(d_{s,t}) = G(d_{\bar{u}}) + B_1(s) + B_2(t),$$

where

$$\begin{aligned} B_1(s) &= \int_{\text{supp } v} \left[g_z(d_{\bar{u}}(z)e^{sv(z)}) - g_z(d_{\bar{u}}(z)) \right] \eta(dz), \\ B_2(t) &= \int_{\text{supp } w} \left[g_z(d_{\bar{u}}(z)e^{tw(z)}) - g_z(d_{\bar{u}}(z)) \right] \eta(dz). \end{aligned}$$

The functions A_1 and A_2 are C^1 , and

$$\begin{aligned} A'_1(s) &= \int_{\text{supp } v} d_{\bar{u}}(z)e^{sv(z)}v(z)f'_z(d_{\bar{u}}(z)e^{sv(z)}) \mu(dz) > 0, \\ A'_2(t) &= \int_{\text{supp } w} d_{\bar{u}}(z)e^{tw(z)}w(z)f'_z(d_{\bar{u}}(z)e^{tw(z)}) \mu(dz) > 0, \end{aligned}$$

for s, t sufficiently close to zero. The strict inequalities follow from $v, w \geq 0$, from the fact that each is positive on a nonempty open set, from full support of μ , and from $f'_z > 0$ μ -almost everywhere. Hence A_1 and A_2 map small neighborhoods of zero homeomorphically onto nondegenerate open intervals.

Using $G = h \circ F$, we obtain

$$G(d_{\bar{u}}) + B_1(s) + B_2(t) = h(F(d_{\bar{u}}) + A_1(s) + A_2(t)).$$

Put $x = A_1(s)$, $y = A_2(t)$, and define

$$\tilde{B}_1(x) = B_1(A_1^{-1}(x)), \quad \tilde{B}_2(y) = B_2(A_2^{-1}(y)).$$

Then, on a rectangle of open intervals,

$$\tilde{B}_1(x) + \tilde{B}_2(y) + G(d_{\bar{u}}) = h(F(d_{\bar{u}}) + x + y).$$

This is exactly of the form of Equation 5 of Lemma A.4. Since h is strictly increasing, the lemma implies that there are constants $A_{\bar{u}} > 0$ and $B_{\bar{u}}$ such that

$$h(r) = A_{\bar{u}}r + B_{\bar{u}}$$

for all r in a neighborhood of $F(d_{\bar{u}})$.

The profile $d_{\bar{u}}$ was arbitrary. Therefore h is locally affine at every point of the interval $\mathcal{I} = F(\mathcal{U}_{\varepsilon}(O))$. Since $\mathcal{I}$ is connected, these local affine expressions must agree on overlaps. Hence there are constants $A > 0$ and $B \in \mathbb{R}$ such that

$$h(r) = Ar + B \quad \text{for every } r \in \mathcal{I}.$$

Consequently,

$$G(d_u) = AF(d_u) + B \quad \text{for every } d_u \in \mathcal{U}_\varepsilon(O).$$

It remains to identify the marginal measures. Fix $d_u \in \mathcal{U}_\varepsilon(O)$ and $\xi \in C_c^2(O)$. For $|r|$ sufficiently small, $d_{u+r\xi} = d_u e^{r\xi}$ remains in $\mathcal{U}_\varepsilon(O)$. Differentiating

$$G(d_{u+r\xi}) = AF(d_{u+r\xi}) + B$$

at $r = 0$, and using the domination assumptions to differentiate under the integral sign, gives

$$\int_O \xi(z) d_u(z) g'_z(d_u(z)) \eta(dz) = A \int_O \xi(z) d_u(z) f'_z(d_u(z)) \mu(dz).$$

Since $d_u(z) > 0$ and since $C_c^2(O)$ test functions determine locally finite Borel measures on O , it follows that

$$g'_z(d_u(z)) \eta(dz) = A f'_z(d_u(z)) \mu(dz)$$

as locally finite Borel measures on O . □

Step 3. Determinant-one unit changes eliminate ray-specific weights.

We first extend Lemma A.1 to all determinant-one unit changes. By conjugating the pairwise rescaling in Lemma A.1 with permutations, the equality

$$W(\lambda x, \lambda R) = W(x, R)$$

holds whenever $\lambda \gg 0$ rescales one good by t , another by t^{-1} , and leaves all remaining goods unchanged. Every $\lambda \gg 0$ with $\prod_{\ell=1}^L \lambda_\ell = 1$ is a finite product of such pairwise rescalings. Therefore

$$W(\lambda x, \lambda R) = W(x, R) \quad \text{whenever } \prod_{\ell=1}^L \lambda_\ell = 1.$$

Write $\tau_\ell = \log \lambda_\ell$. Then $\tau \in H$, and

$$\lambda a(z) = a(z + \tau).$$

Hence the transformed ray profile is

$$(T_\tau d)(z) = d(z - \tau).$$

Since F represents the ordering induced by W , equality of W -values implies equality of F -values. Thus

$$F(T_\tau d) = F(d)$$

for every attainable profile d .

Now

$$F(T_\tau d) = \int_H \phi_z(d(z - \tau)) \nu(dz).$$

After the change of variables $z \mapsto z + \tau$, this becomes

$$F(T_\tau d) = \int_H \phi_{z+\tau}(d(z)) \nu^\tau(dz),$$

where

$$\nu^\tau(B) = \nu(B + \tau)$$

for every Borel set $B \subset H$.

Fix $a > 0$, and put

$$d^a(z) = ab(z).$$

Let $K \subset H$ be compact. By Lemma A.3, for all sufficiently small $u \in C_c^2(K)$, the profile

$$d_u(z) = d^a(z)e^{u(z)}$$

is attainable. Hence

$$\int_H \phi_{z+\tau}(d_u(z)) \nu^\tau(dz) = \int_H \phi_z(d_u(z)) \nu(dz).$$

Differentiate at $u = 0$ in an arbitrary direction $\xi \in C_c^2(K)$. Since

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} d^a(z)e^{\varepsilon\xi(z)} = d^a(z)\xi(z),$$

we obtain

$$\int_K \xi(z) d^a(z) \phi'_{z+\tau}(d^a(z)) \nu^\tau(dz) = \int_K \xi(z) d^a(z) \phi'_z(d^a(z)) \nu(dz).$$

Because $d^a(z) > 0$ and ξ is arbitrary,

$$\phi'_z(ab(z)) \nu(dz) = \phi'_{z+\tau}(ab(z)) \nu^\tau(dz) \tag{1}$$

as measures on K . Since K was arbitrary, (1) holds on H . Notice that this is still only an identity along the graph $t = ab(z)$. The following patching lemma converts it into a fixed-level identity.

Lemma A.6 (Patching graph identities). *Let μ_1, μ_2 be positive locally finite Borel measures on H . Let $p_z, q_z : \mathbb{R}_{++} \rightarrow \mathbb{R}_{++}$ be C^1 in t for almost every z , and assume that, for every compact $K \subset H$ and every compact interval $J \Subset \mathbb{R}_{++}$,*

$$\int_K \sup_{s \in J} |\partial_t p_z(s)| \mu_1(dz) + \int_K \sup_{s \in J} |\partial_t q_z(s)| \mu_2(dz) < \infty.$$

Let $b : H \rightarrow \mathbb{R}_{++}$ be continuous. Suppose that, for every $a > 0$,

$$p_z(ab(z)) \mu_1(dz) = q_z(ab(z)) \mu_2(dz)$$

as measures on H . Then, for every fixed $t > 0$,

$$p_z(t) \mu_1(dz) = q_z(t) \mu_2(dz)$$

as measures on H .

Proof. Fix $t > 0$, a compact set $K \subset H$, and a Borel set $B \subset K$. Since b is positive and continuous, $\log b$ is uniformly continuous on K . For $\delta > 0$, choose a finite Borel partition $K = E_1 \cup \dots \cup E_N$ and points $z_i \in E_i$ such that

$$|\log b(z) - \log b(z_i)| < \delta \quad \text{for all } z \in E_i.$$

Put

$$a_i = \frac{t}{b(z_i)}.$$

Then

$$a_i b(z) \in [te^{-\delta}, te^{\delta}] \quad \text{for all } z \in E_i.$$

Let $J_\delta = [te^{-\delta}, te^{\delta}]$. By the assumed graph identity,

$$\int_{B \cap E_i} p_z(a_i b(z)) \mu_1(dz) = \int_{B \cap E_i} q_z(a_i b(z)) \mu_2(dz).$$

Therefore

$$\begin{aligned} & \left| \int_B p_z(t) \mu_1(dz) - \int_B q_z(t) \mu_2(dz) \right| \\ & \leq \sum_i \int_{B \cap E_i} |p_z(t) - p_z(a_i b(z))| \mu_1(dz) \\ & \quad + \sum_i \int_{B \cap E_i} |q_z(t) - q_z(a_i b(z))| \mu_2(dz) \\ & \leq \omega_\delta \left[\int_K \sup_{s \in J_\delta} |\partial_t p_z(s)| \mu_1(dz) + \int_K \sup_{s \in J_\delta} |\partial_t q_z(s)| \mu_2(dz) \right], \end{aligned}$$

where

$$\omega_\delta = \sup_i \sup_{z \in E_i} |a_i b(z) - t|.$$

As $\delta \downarrow 0$, $\omega_\delta \rightarrow 0$. For δ small, the intervals J_δ are contained in a fixed compact subinterval of $\mathbb{R}_{++}$, so the bracketed quantity is finite by assumption. Hence the difference above is zero. Since $B \subset K$ and $K \subset H$ were arbitrary, the fixed-level identity holds on H . $\square$

Apply Lemma A.6 to

$$p_z(t) = \phi'_z(t), \quad q_z(t) = \phi'_{z+\tau}(t), \quad \mu_1 = \nu, \quad \mu_2 = \nu^\tau.$$

The regularity condition for q follows from the regularity condition for ϕ by translation. From (1), we get, for every $t > 0$,

$$\phi'_z(t) \nu(dz) = \phi'_{z+\tau}(t) \nu^\tau(dz). \quad (2)$$

Define the marginal measure

$$M_t(dz) = \phi'_z(t) \nu(dz).$$

Then (2) implies

$$M_t(B) = M_t(B + \tau)$$

for every Borel set $B \subset H$ and every $\tau \in H$. Thus M_t is a locally finite translation-invariant Borel measure on the finite-dimensional vector space H . By uniqueness of Haar measure, there exists $k(t) > 0$ such that

$$M_t(dz) = k(t) ds(z). \quad (3)$$

The function k is continuous: for any compact $Q \subset H$ with $0 < s(Q) < \infty$,

$$k(t) = \frac{1}{s(Q)} \int_Q \phi'_z(t) \nu(dz),$$

and continuity follows from the regularity assumption and dominated convergence.

Step 4. Common unit rescalings force a power marginal.

Let $c > 0$. A common unit rescaling sends a profile d to cd . Unit Invariance implies that

$$F_c(d) := F(cd)$$

represents the same ordering as $F(d)$. Fix $a > 0$ and a compact $K \subset H$. On the local chart around $d^a(z) = ab(z)$, Lemma A.5 gives constants $A = A(c, a, K) > 0$ and $B = B(c, a, K)$ such that

$$F_c = AF + B$$

on that chart. Differentiating at d^a in arbitrary directions $\xi \in C_c^2(K)$ gives

$$c \phi'_z(cab(z)) \nu(dz) = A(c, a, K) \phi'_z(ab(z)) \nu(dz)$$

as measures on K . Using (3), this becomes

$$c k(cab(z)) = A(c, a, K) k(ab(z)) \quad \text{for } s\text{-a.e. } z \in K. \quad (4)$$

Since b and k are continuous, the equality holds everywhere on the interior of K .

For fixed $c > 0$, define

$$R_c(t) = \frac{c k(ct)}{k(t)}.$$

Equation (4) says that $R_c(ab(z))$ is constant on every relatively compact open set. On overlapping open sets the constants must agree. Since H is connected, the constant is independent of the chosen local chart. Therefore, for every fixed $a > 0$, R_c is constant on the range $ab(H)$.

The range of the CES base profile is explicit. If $0 < \rho < 1$, then

$$b(H) = (0, \beta], \quad \beta = L^{-1/\rho}.$$

If $\rho < 0$, then

$$b(H) = [\beta, \infty), \quad \beta = L^{-1/\rho}.$$

In the lower branch, the intervals $ab(H) = (0, a\beta]$ cover $\mathbb{R}_{++}$, and any two positive numbers belong to a common such interval by taking a large enough. In the upper branch, the intervals $ab(H) = [a\beta, \infty)$ cover $\mathbb{R}_{++}$, and any two positive numbers belong to a common such interval

by taking a small enough. Hence, for each $c > 0$, $R_c(t)$ is constant on all of $\mathbb{R}_{++}$. Thus there exists $A(c) > 0$ such that

$$c k(ct) = A(c)k(t) \quad \text{for every } t > 0.$$

Equivalently,

$$k(ct) = B(c)k(t), \quad B(c) = \frac{A(c)}{c}.$$

Setting $t = 1$ gives

$$B(c) = \frac{k(c)}{k(1)}.$$

Therefore

$$k(ct) = \frac{k(c)k(t)}{k(1)} \quad \text{for all } c, t > 0.$$

Since k is positive and continuous, the multiplicative Cauchy equation implies that

$$k(t) = Kt^{p-1} \tag{5}$$

for some $K > 0$ and some $p \in \mathbb{R}$.

The logarithmic case $p = 0$ is impossible. Indeed, from (3) with $t = 1$, ν is equivalent to ds . Write $\nu = w ds$. Then (3) and (5) imply, on a common full-measure set,

$$\phi'_z(t)w(z) = Kt^{p-1} \quad \text{for every } t > 0.$$

If $p = 0$, then for any $c > 1$ and any admissible profile d ,

$$\begin{aligned} F(cd) - F(d) &= \int_H \int_{d(z)}^{cd(z)} \phi'_z(t) dt \nu(dz) \\ &= \int_H \int_{d(z)}^{cd(z)} Kt^{-1} dt ds(z) \\ &= K \log(c) s(H) = +\infty, \end{aligned}$$

because $s(H) = \infty$. This contradicts the fact that F is real-valued on the domain. Hence

$$p \neq 0.$$

For $p \neq 0$, the same calculation gives, for any two admissible profiles d, e for which the displayed integrals are finite,

$$F(d) - F(e) = \frac{K}{p} \left(\int_H d(z)^p ds(z) - \int_H e(z)^p ds(z) \right). \tag{6}$$

Step 5. The exponent belongs to the announced range. Real-valuedness of F implies that

$$\int_H d(z)^p ds(z) < \infty$$

for every admissible profile d in the branch. Indeed, suppose that some admissible profile d satisfied

$$\int_H d(z)^p ds(z) = \infty.$$

Since cd is admissible for every $c > 0$, the pointwise identity following from (5) gives, for $c \neq 1$,

$$F(cd) - F(d) = \frac{K}{p}(c^p - 1) \int_H d(z)^p ds(z),$$

where the right-hand side is an extended real number. This is impossible because F is real-valued on the domain. Hence

$$\int_H d(z)^p ds(z) < \infty$$

for every admissible profile d .

Now use the sharpness of the robust volume branches at their boundary exponent. On the lower branch, put

$$q_* = L\bar{\alpha}.$$

Since in part (i) we have $\bar{\alpha} \geq 0$, $q_* \geq 0$. The domain $\mathcal{R}_{\bar{\alpha}}^+$ contains smooth admissible profiles satisfying

$$\int_H d(z)^q ds(z) < \infty \iff q > q_*.$$

For $q_* > 0$, such profiles are obtained from smooth homogeneous convex contours whose radial functions are comparable at infinity to

$$d(z) \asymp (1 + \|z\|)^{-(L-1)/q_*}.$$

Since H has dimension $L - 1$, this tail gives

$$\int_H d(z)^q ds(z) < \infty \iff q > q_*.$$

The case $q_* = 0$ is obtained from the CES tails used above, for which

$$\int_H d(z)^q ds(z) < \infty \iff q > 0.$$

These profiles belong to $\mathcal{R}_{\bar{\alpha}}^+$, because the lower robust volume condition requires finiteness only for exponents

$$q = L\alpha > L\bar{\alpha} = q_*.$$

Applying the conclusion already proved, namely finiteness of $\int_H d(z)^p ds(z)$ for every admissible profile, to such a boundary-sharp profile yields

$$p > q_* = L\bar{\alpha}.$$

In particular $p > 0$ on the lower branch.

On the upper branch, put again

$$q_* = L\bar{\alpha}.$$

Since in part (ii) we have $\bar{\alpha} \leq 0$, $q_* \leq 0$. The domain $\mathcal{R}_{\bar{\alpha}}^-$ contains smooth admissible profiles satisfying

$$\int_H d(z)^q ds(z) < \infty \iff q < q_*.$$

For $q_* < 0$, such profiles are obtained from smooth homogeneous convex contours whose radial functions are comparable at infinity to

$$d(z) \asymp (1 + \|z\|)^{(L-1)/|q_*|}.$$

Since H has dimension $L - 1$, this tail gives

$$\int_H d(z)^q ds(z) < \infty \iff q < q_*.$$

The case $q_* = 0$ is obtained from the upper CES tails, for which

$$\int_H d(z)^q ds(z) < \infty \iff q < 0.$$

These profiles belong to $\mathcal{R}_{\bar{\alpha}}^-$, because the upper robust volume condition requires finiteness only for exponents

$$q = L\alpha < L\bar{\alpha} = q_*.$$

Applying finiteness of $\int_H d(z)^p ds(z)$ for every admissible profile to such a boundary-sharp profile yields

$$p < q_* = L\bar{\alpha}.$$

In particular $p < 0$ on the upper branch.

Put

$$\alpha = \frac{p}{L}.$$

Step 6. Representation on the lower-volume branch.

Assume $\bar{\alpha} \geq 0$. By Step 5,

$$p > L\bar{\alpha}, \quad \alpha = \frac{p}{L} > \bar{\alpha}.$$

Since $p > 0$, (6) shows that F , and therefore W , is ordinally equivalent to

$$d \mapsto \int_H d(z)^p ds(z).$$

By Proposition 5.4, for $p = \alpha L$,

$$\text{Vol}_\alpha(L(x, R)) = C_{\alpha, L} \int_H d_{x, R}(z)^p ds(z)$$

for some constant $C_{\alpha, L} > 0$, whenever the right-hand side is finite. Therefore, on $X_{++} \times \mathcal{R}_{\bar{\alpha}}^+$,

$$W \sim \text{Vol}_\alpha(L(x, R)).$$

Because $\alpha > 0$, the map $v \mapsto v^{1/\alpha}$ is strictly increasing. Hence

$$W \sim \left(\text{Vol}_\alpha(L(x, R)) \right)^{1/\alpha}.$$

It remains to extend the representation to boundary bundles. Define

$$G_+(x, R) = (\text{Vol}_\alpha(L(x, R)))^{1/\alpha}.$$

The converse argument above has shown that, for each fixed $R \in \mathcal{R}_\alpha^+$, the map $x \mapsto G_+(x, R)$ is continuous on all of X . We have proved that W and G_+ induce the same ordering on $X_{++} \times \mathcal{R}_\alpha^+$. Since X_{++} is dense in X and both $W(\cdot, R)$ and $G_+(\cdot, R)$ are continuous for each fixed R , this ordering extends uniquely to boundary situations. Hence

$$W \sim G_+$$

on all of $X \times \mathcal{R}_\alpha^+$. This proves part (i).

Step 7. Representation on the upper-volume branch.

Assume $\bar{\alpha} \leq 0$. By Step 5,

$$p < L\bar{\alpha}, \quad \alpha = \frac{p}{L} < \bar{\alpha}.$$

Since $p < 0$, (6) shows that F , and therefore W , is ordinally equivalent to

$$d \mapsto - \int_H d(z)^p ds(z).$$

By Proposition 5.4, for $p = \alpha L$,

$$\text{Vol}_\alpha(U(x, R)) = C_{\alpha, L} \int_H d_{x, R}(z)^p ds(z)$$

for some constant $C_{\alpha, L} > 0$, whenever $x \in X_{++}$. Therefore

$$W \sim - \text{Vol}_\alpha(U(x, R)).$$

Because $\alpha < 0$, the map $v \mapsto v^{1/\alpha}$ is strictly decreasing. Thus ranking by $-\text{Vol}_\alpha(U(x, R))$ is equivalent to ranking by

$$\left(\text{Vol}_\alpha(U(x, R)) \right)^{1/\alpha}.$$

Hence, on $X_{++} \times \mathcal{R}_\alpha^-$,

$$W \sim \left(\text{Vol}_\alpha(U(x, R)) \right)^{1/\alpha}.$$

If $x \in \partial X$, then the upper contour touches the boundary of the orthant. Since $\alpha < 0$, the density $\prod_{\ell=1}^L y_\ell^{\alpha-1}$ is not integrable near the boundary, and therefore

$$\text{Vol}_\alpha(U(x, R)) = +\infty.$$

Thus

$$\left(\text{Vol}_\alpha(U(x, R)) \right)^{1/\alpha} = 0,$$

because $1/\alpha < 0$. Continuity pins down this boundary value. Hence

$$W \sim \begin{cases} \left(\text{Vol}_\alpha(U(x, R))\right)^{1/\alpha}, & x \in X_{++}, \\ 0, & x \in \partial X. \end{cases}$$

This proves part (ii).

□